NetVis: A Composable Architecture for Large-Scale Network Visualization

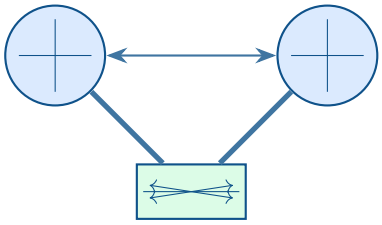
Technical Reference

Simon Knight, Ph.D.

Adelaide, Australia

March 2026 • Version 1.5.0

Last updated: March 11, 2026



Abstract. NetVis is a Rust-based network topology layout and visualization engine comprising 90,000+ lines of code. It transforms complex multi-layer network topologies into publication-quality visualizations using composable layout algorithms, R*-tree–accelerated label placement, force-directed edge bundling, and adaptive level-of-detail rendering. This document is the definitive technical reference for NetVis. It describes the data model, architecture, eight layout engines, ten edge routing and post-layout refiners, parallel edge rendering (five modes with automatic LAG/bond aggregation), interface rendering, node shape library, edge gradients, path emphasis, metric encoding, rendering pipeline, topology diffing, temporal analysis, traffic animation, annotation markup, interactive editing, visual effects, graph analytics, filter DSL, 16 quality metric families, and public API. Appendices provide the full YAML topology schema, performance baselines with comparative benchmarks against Graphviz, and a complete configuration reference.

Contents

LAG Link Aggregation Group . . . 7
DSL Domain-Specific Language . . . 30
LOD Level of Detail . . . 10

1 Introduction

NetVis is a network topology layout and visualization engine. Given a graph of network devices and connections (from a YAML file, NetBox [1] instance, or LLDP discovery), it produces publication-quality SVG diagrams, interactive HTML visualizations, or PNG/PDF exports. It addresses the gap between general-purpose graph layout tools (Graphviz [2], OGDF [3], D3.js [4]) and manual diagramming applications (yEd [5], Visio).

1.1 Design Philosophy

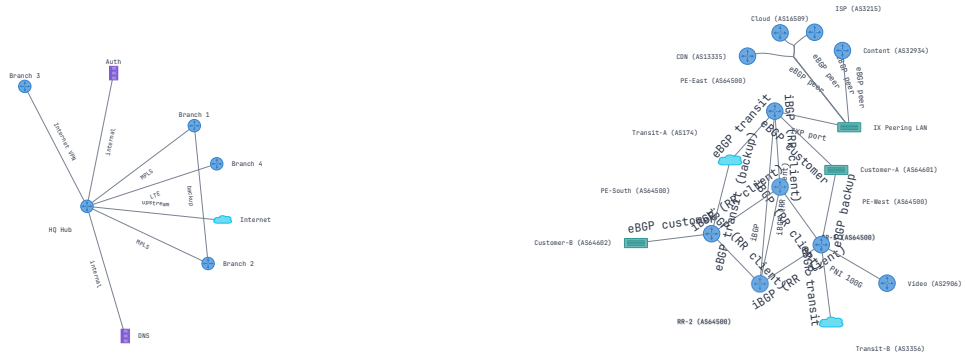
Two principles guide every design decision in NetVis:

Domain-specialized composition. Network topologies have exploitable structural regularity—hierarchical tiers, geographic site placement, protocol-layer stacking. Rather than using a single general-purpose layout algorithm, NetVis decomposes visualization into a pipeline of composable refiners, each specialized for one aspect of the output.

Quality by default. The system should produce readable, labeled, styled diagrams without manual post-processing. Quality metrics (crossings, overlaps, label readability) are measured as part of the rendering pipeline, not as an afterthought.

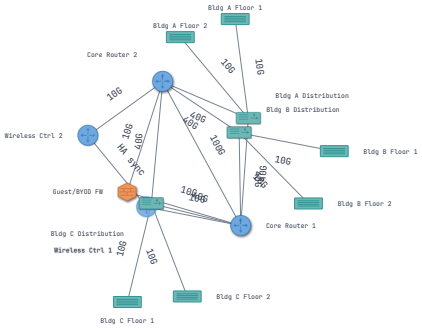
1.2 Output Gallery

Figures 1 through 3 show representative outputs across different layout modes and topology scales. Every diagram was generated automatically from YAML with no manual adjustment.



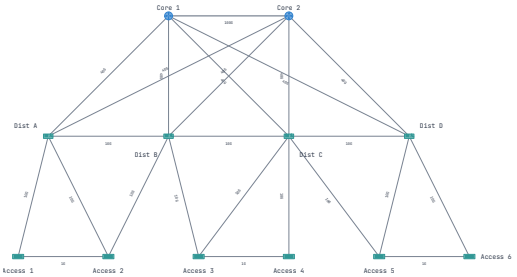
(a) WAN hub-and-spoke (radial tree, 18 nodes) **(b)** BGP peering fabric (force-directed, 24 nodes)

Figure 1. Representative NetVis outputs. All diagrams generated automatically from YAML with no manual adjustment.

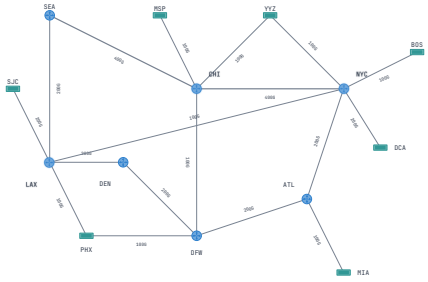


(a) Enterprise campus (real-world topology, 72 nodes) **(b)** Datacenter spine-leaf (real-world, 96 nodes)

Figure 2. Real-world topology examples imported directly from YAML exports.



(a) Edge bundling showcase (FDEB, 200 edges)



(b) Subway layout (octilinear, ring backbone)

Figure 3. Specialized layout modes. Left: FDEB reduces clutter by grouping spatially compatible edges. Right: subway layout constrains all edges to 0/45/90 degree routes for transit-map clarity.

1.3 Audience

This document targets three audiences: engineers embedding NetVis as a library in a network management platform (via Rust, Python, or WASM); contributors to the NetVis codebase; and researchers who want to understand the algorithms in detail. Familiarity with basic graph theory is assumed; familiarity with network engineering concepts (spine-leaf, OSPF, BGP) is helpful but not required.

1.4 How to Read This Document

- **Quick start:** Read §2 (By Example), then §27 (API and Usage).
- **Data model and architecture:** §3–§4.
- **Deep dive into algorithms:** §5–§24.
- **Analysis features:** §15–§19 (diff, timeline, traffic, annotations, editor).
- **Deployment and configuration:** §28 and Appendix B.
- **Known issues:** §31.

2 NetVis by Example

This section walks through representative capabilities using minimal YAML examples. Full schema details appear in Appendix A; CLI options in Appendix B.

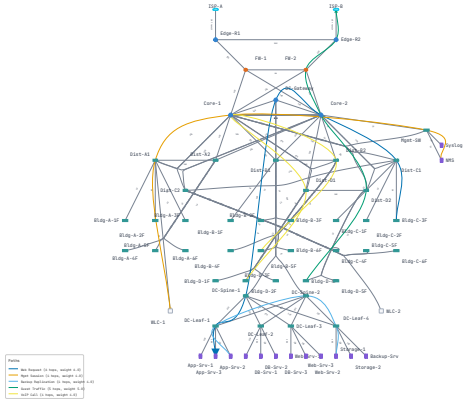
2.1 Basic Campus Network

The minimal topology YAML requires only `name`, `nodes`, and `edges`. The rendering section is optional; NetVis selects a suitable layout automatically:

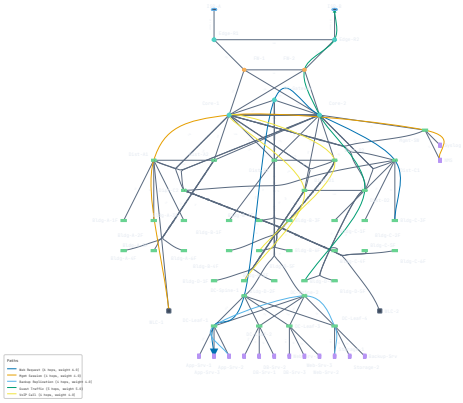
Minimal topology (campus-edge.yaml)

```
name: Campus edge
render:
  theme: light
nodes:
  - { id: inet, label: Internet, node_type: cloud }
  - { id: fw1,  label: Firewall, node_type: firewall }
  - { id: r1,   label: Edge Router, node_type: router }
  - { id: s1,   label: Access Switch, node_type: switch }
edges:
  - { from: inet, to: fw1,  label: uplink }
  - { from: fw1,  to: r1,   label: transit }
  - { from: r1,   to: s1,   label: 10G }
```

```
netvis –input campus-edge.yaml -o campus.svg
```



(a) Enterprise campus, light theme

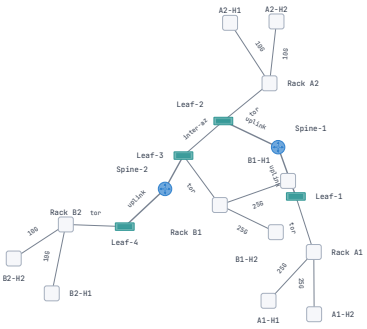


(b) Same topology, dark theme

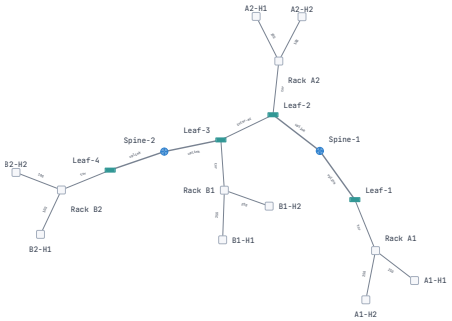
Figure 4. The same enterprise campus topology in two themes. The YAML is identical; only `theme:` changes. See §6 for the full theme gallery.

2.2 Layout Algorithm Comparison

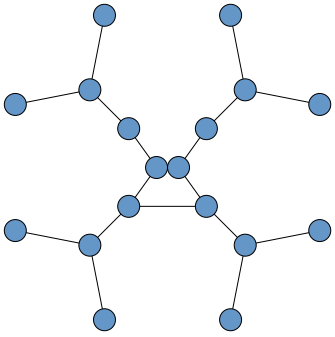
The `preset` or `layout` field selects the layout engine. Figure 5 shows the same datacenter topology rendered with four different engines.



(a) Force-directed (default)



(b) Hierarchical (Sugiyama)



(c) Radial tree

(d) Spine-leaf preset

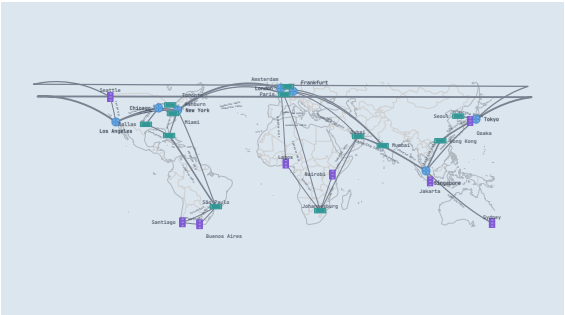
Figure 5. The same 48-node datacenter topology rendered with four layout algorithms. Hierarchical most clearly exposes the two-tier structure; radial is best for hub-centric views; force-directed and spine-leaf are both suitable for ad-hoc exploration.

2.3 Geographic CDN

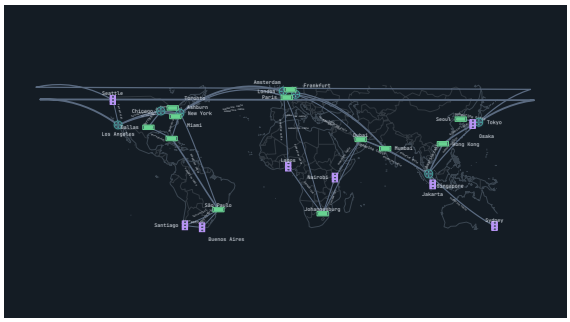
Nodes with `lat`/`lon` attributes use the geographic layout with great-circle arc edges and a background map layer:

Geographic layout (cdn.yaml, excerpt)

```
render:
  preset: geographic
  theme: dark
geographic:
  show_map: true
  great_circle_arcs: true
nodes:
  - { id: nyc, label: New York, node_type: router,
      lat: 40.7128, lon: -74.0060 }
  - { id: lon, label: London, node_type: router,
      lat: 51.5074, lon: -0.1278 }
```



(a) Geographic layout, light theme



(b) Geographic layout, dark theme

Figure 6. Global CDN in geographic layout. Great-circle arc edges and background map layer. The dark theme is preferred for geographic diagrams as map features remain legible.

2.4 Isometric Multi-Layer

The isometric layout renders protocol stacks or layered architectures with a pseudo-3D perspective. Each layer runs an independent force-directed sub-layout:

Isometric layout (multilayer.yaml, excerpt)

```
render:
  layout: isometric
layers:
  - { id: physical, label: Physical, order: 0 }
  - { id: network,  label: Network,  order: 1 }
  - { id: routing,  label: Routing,  order: 2 }
nodes:
  - { id: p-core1, label: Core-1, node_type: router,
      attributes: { layer: physical } }
```

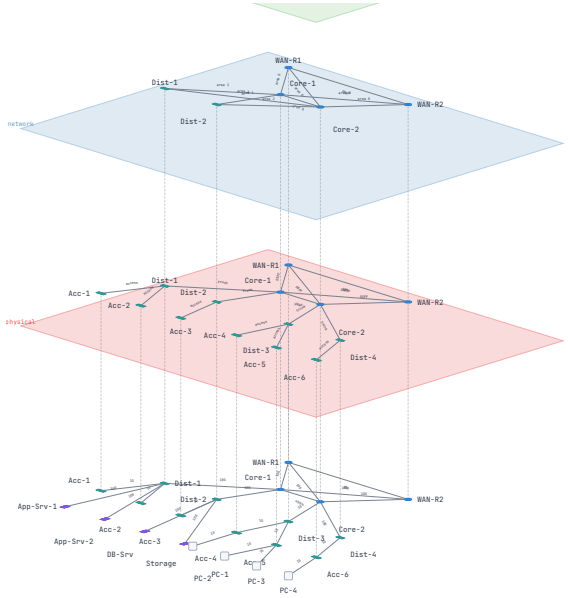


Figure 7. Isometric multi-layer: three protocol planes (Physical, Network, Routing) separated along the Z-axis. Layer planes render as filled SVG polygons. The projection angle is $\phi = 18^{\circ}$ with 350 scene units of layer spacing.

2.5 Parallel Edges and Bond Groups

Multiple physical links between the same pair of nodes are grouped into an `EdgeGroup`. The `parallel_mode` field selects the rendering mode; bond groups aggregate bandwidth automatically:

Parallel edge modes

```
edges:
  # LAG: 4x 10G auto-aggregated to 40G
  - { from: core1, to: core2, label: "10G",
      bond_group: core-lag, parallel_mode: bundled }
  - { from: core1, to: core2, label: "10G",
      bond_group: core-lag }
  # Redundant uplinks shown as offset lines
  - { from: core1, to: dist1, label: "10G",
      parallel_mode: offset }
  - { from: core1, to: dist1, label: "10G" }
```

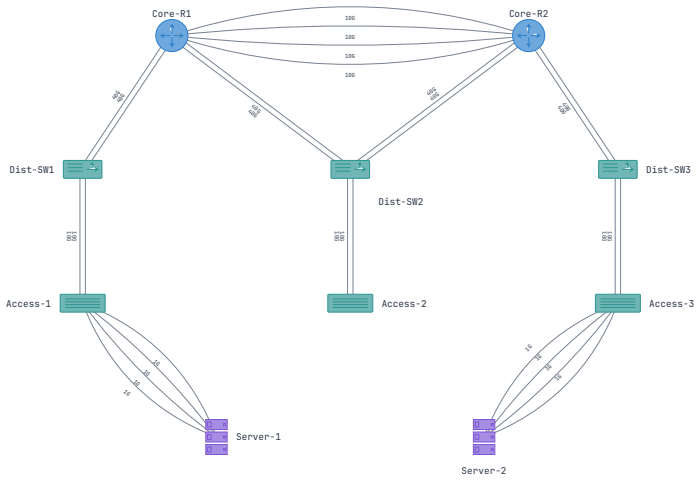


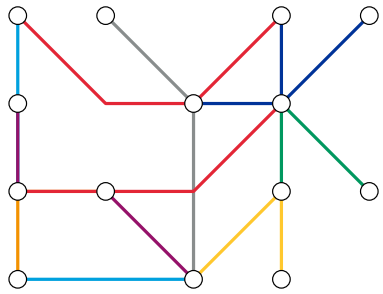
Figure 8. Parallel edge rendering modes on a datacenter topology. The diagram demonstrates Bundled (core Link Aggregation Group (LAG)), Offset (redundant uplinks), BondingIcon (bonded access), and Stacked (server NICs) modes. See §7 for details.

2.6 Subway Layout

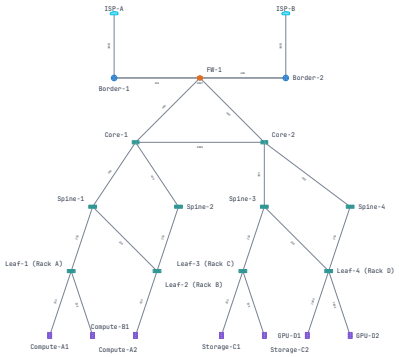
The subway layout constrains all edges to 0, 45, or 90 degree routes and snaps nodes to a uniform grid. The blueprint theme complements the geometric precision:

Subway layout

```
render:
  layout: subway
  theme: blueprint
```



(a) Subway layout, blueprint theme



(b) Orthogonal layout

Figure 9. Octilinear and orthogonal layouts. Both enforce geometric discipline: subway at 0/45/90 degrees; orthogonal at strict Manhattan-style horizontal and vertical segments.

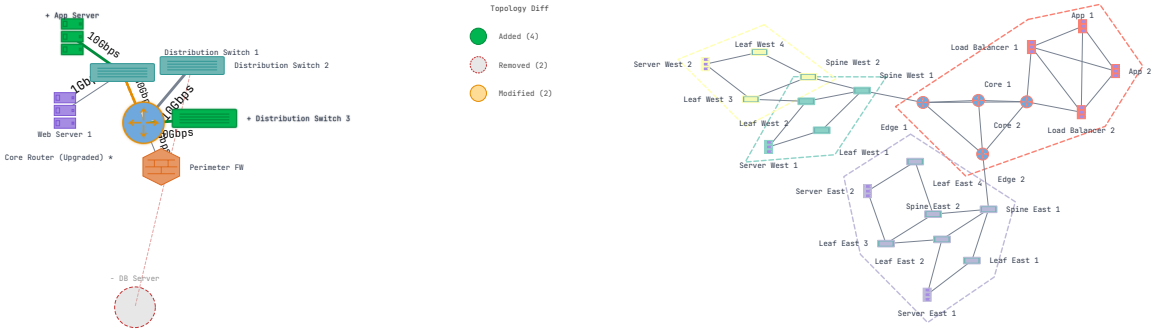
2.7 Topology Diff and Analysis

NetVis can structurally compare two topologies and produce a visual diff (added in green, removed in red, modified in amber) and compute graph analytics:

Topology diff and analytics

```
# Visual diff between topology versions
netvis diff old.yaml new.yaml -o diff.svg
```

```
# Graph analytics overlay
netvis --input topology.yaml --analytics -o analytics.svg
```



(a) Topology diff: green=added, red=removed, amber=modified

(b) Community detection with group boundaries

Figure 10. Analysis features. Left: topology diff visualizes structural changes between two versions. Right: automatic community detection partitions nodes into groups with colored boundary overlays.

3 Data Model

NetVis uses a directed graph as its core data structure, built on the `petgraph` library [6]. The core type, `NetVisGraph`, is a `StableGraph<NodeData, EdgeData>`; the `StableGraph` variant is used because node and edge indices remain valid after removal—important for the editor's undo/redo system.

3.1 Node and Edge Data

Field	Type	Purpose
NodeData		
`id`	String	Unique identifier (e.g., `"router-fra-01"`)
`label`	Option<String>	Display text
`node_type`	Option<String>	Device type driving icon and default styling
`attributes`	HashMap	Arbitrary metadata for metric encoding and filter DSL
`position`	Option<Point2D>	Layout hint; overrides engine if set
`group`	Option<GroupId>	Group membership (community detection, composite)
`layer`	Option<LayerId>	Isometric layer membership
EdgeData		
`label`	Option<String>	Edge label
`edge_type`	Option<String>	Semantic type: p2p, trunk, submarine, …
`directed`	bool	Default `false`
`bond_group`	Option<String>	LAG/bond group identifier
`weight`	f64	Default 1.0; used as spring constant in force-directed
`attributes`	HashMap	Arbitrary metadata

Table 1. Core graph data fields.

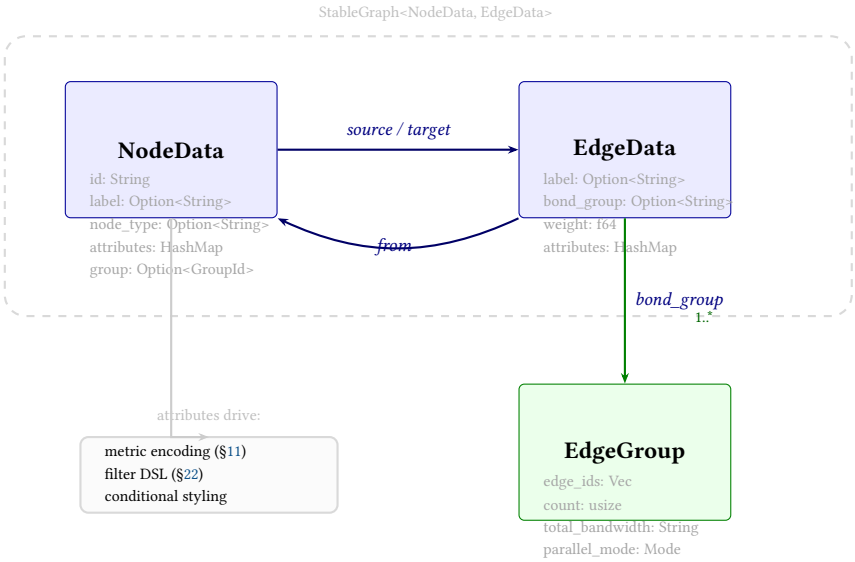


Figure 11. Data model entity relationships. `NodeData` and `EdgeData` live in a `petgraph::StableGraph` whose indices remain stable across removals. Edges sharing a `bond_group` are aggregated into an `EdgeGroup` with computed bandwidth totals. The `attributes` HashMap drives metric encoding, filter DSL evaluation, and conditional styling.

3.2 Edge Groups

When multiple edges connect the same pair of nodes, NetVis groups them into an `EdgeGroup` that carries aggregate metadata: the member `edge_ids`, a `count`, an automatically computed `total_bandwidth` (e.g., four edges labeled `"10G"` produce `"40G"`), and a `parallel_mode` (see §7).

> **Note**
>
> The `node_type` field drives device-specific rendering: icon shape, default color, and default size. The full set of built-in node types is described in §8. Unknown values render as circles.

3.3 Scene Graph

Layout engines consume a `NetVisGraph` and produce a `Scene`: a flat list of positioned nodes (with shapes, colors, labels), edges (as Bézier paths), label anchors, group boundaries, and layer planes. Refiners transform the scene between layout and rendering. The scene at 5,000 nodes consumes approximately 9.7 MB of heap.

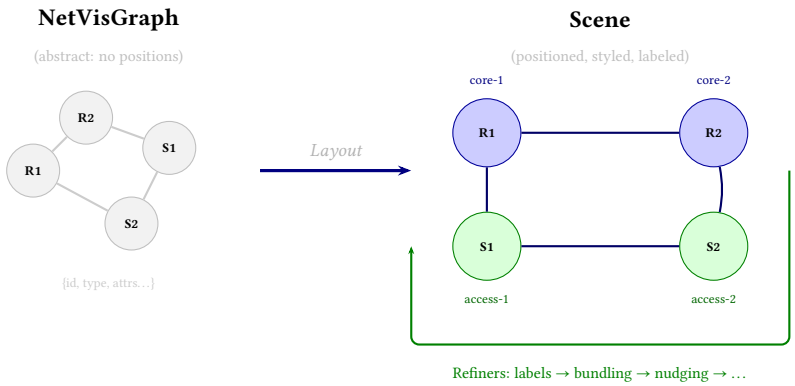


Figure 12. Graph-to-scene transformation. The abstract `NetVisGraph` (left) carries only topology structure. A layout engine assigns positions, shapes, and colors to produce a `Scene` (right). Refiners iteratively transform the scene—placing labels, bundling edges, nudging overlaps—before the renderer draws it.

4 Architecture

NetVis is organized as a six-layer pipeline.

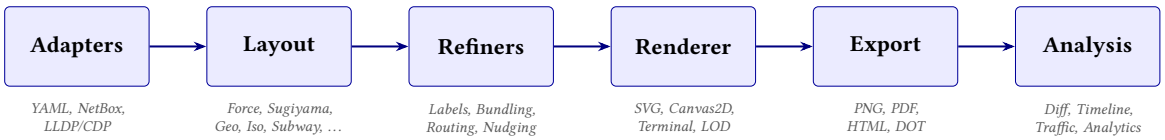


Figure 13. The NetVis six-layer pipeline.

The pipeline stages are:

1. **Adapters.** Stateless converters that ingest topology data from YAML, NetBox [1] REST API, or LLDP/CDP JSON, each implementing `fetch`/`validate`/`to_graph`.
2. **Layout engines.** Pluggable algorithms behind a common `Layout` trait. A layout takes a `NetVisGraph` and returns a `Scene`. Eight engines ship: force-directed (Standard and Enterprise), hierarchical (Sugiyama [7]), radial tree, geographic (Mercator), isometric multi-layer, orthogonal, subway (octilinear), tree, and composite (multi-region).
3. **Refiners.** Post-layout transformations. Current refiners include label placement (R*-tree–accelerated), edge bundling (FDEB), constraint solving, overlap removal, edge nudging, label nudging, edge clipping, smart label snapping, starburst grouping, boundary bundling, port docking, and orthogonal edge routing.
4. **Renderers.** SVG (primary), Canvas2D (browser), and ASCII terminal (`netvis show`). All support Level of Detail (LOD) suppression and viewport culling.
5. **Export.** SVG is optionally converted to PNG (via `resvg`), PDF (via `svg2pdf`), DOT (Graphviz interoperability), or self-contained interactive HTML.
6. **Analysis.** Post-render modules: topology diffing, temporal snapshots, traffic flow animation, SVG annotation markup, and graph analytics.

4.1 Pipeline Composition

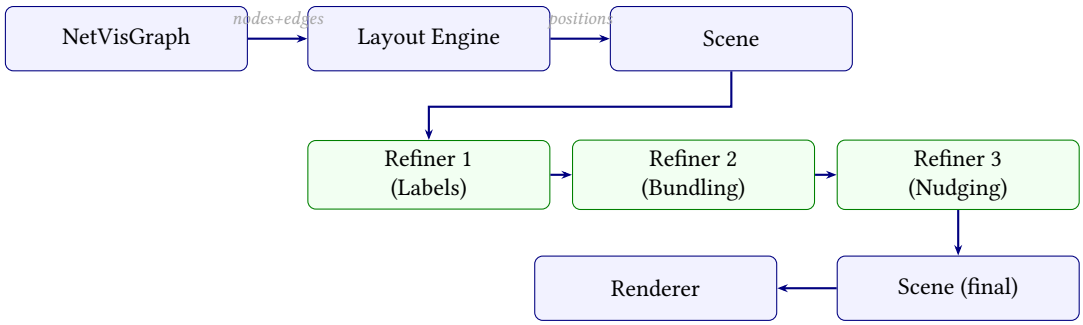


Figure 14. Pipeline composition: layout engine produces a `Scene`, which flows through a chain of refiners before reaching the renderer. Each refiner implements `trait LayoutRefiner` and may add, remove, or reposition scene elements without knowledge of the other refiners.

In YAML, the refiner chain is expressed as:

Pipeline composition in YAML

```
render:
  layout: force_directed
  refiners: [labels, bundling, nudging, constraints]
```

The trait boundary prevents cross-component optimization (e.g., bundling cannot influence label placement during the same pass). This is mitigated by `reposition_edge_labels()`, which re-places labels after bundling detects significant curvature changes.

4.2 Scene Graph as Intermediate Representation

The `Scene` struct carries fully positioned nodes (shapes, colors, labels) and edges (Bézier paths). This separation of concerns means renderers are simple—they draw exactly what the scene contains—at the cost of carrying all rendering data in memory. The scene at 5,000 nodes consumes approximately 9.7 MB of heap.

5 Layout Algorithms

NetVis ships eight layout engines, each implementing the `Layout` trait. The algorithm is selected via the `layout` or `preset` YAML field, or the `--layout`/`--preset` CLI flags.

5.1 Force-Directed Layout

Position nodes in 2D such that connected nodes are close and unconnected nodes are separated, following the spring-electrical model of Fruchterman and Reingold [8]. NetVis ships two variants and selects between them automatically: the **Standard engine** ($O(n^2)$, wraps the `fjadra` library with Verlet integration [9]) for graphs under 2,000 nodes; the **Enterprise engine** (Barnes-Hut $O(n \log n)$ [10], custom implementation with opening angle $\theta = 0.9$) for larger graphs. On non-WASM targets, force calculation is parallelized via Rayon.

Key parameters: velocity damping $\gamma = 0.6$; alpha decay $1 - 0.001^{1/300}$; iteration count $\min(300, 50 + \lfloor n/10 \rfloor)$; minimum edge length 60 pt (enforced post-simulation).

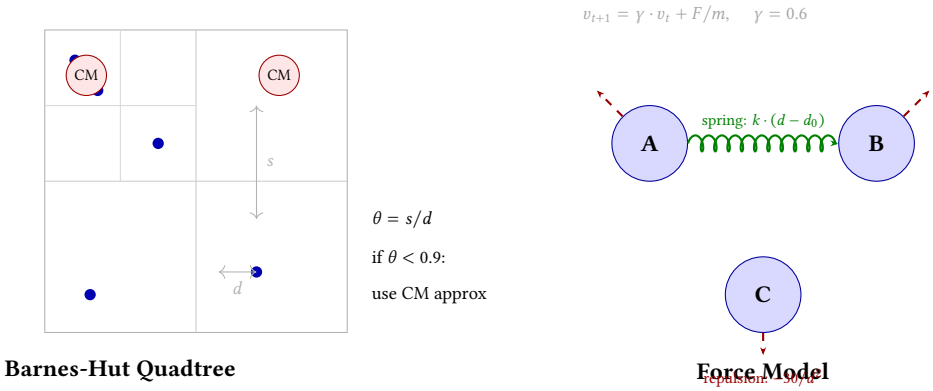


Figure 15. Force-directed layout internals. **Left:** Barnes-Hut quadtree ($O(n \log n)$) approximates distant node groups as a single center-of-mass (CM) when opening angle $\theta = s/d < 0.9$. **Right:** each node experiences attractive spring forces along edges and repulsive charge forces from all other nodes, with velocity damping $\gamma = 0.6$.

5.2 Hierarchical Layout (Sugiyama)

For DAG-structured or tiered networks (e.g., spine-leaf datacenters). The four-phase Sugiyama framework [7]: (1) cycle removal by edge reversal; (2) longest-path-first layer assignment; (3) crossing minimization via barycenter or median heuristic; (4) coordinate assignment to minimize edge bends. Performance: 19 ms at 1,000 nodes.

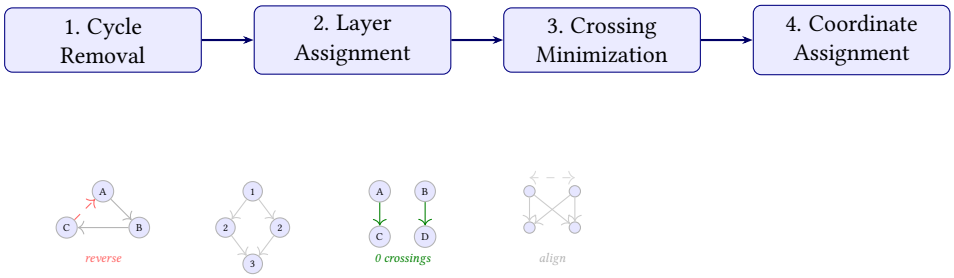


Figure 16. Sugiyama four-phase pipeline. Each phase progressively refines the layout: cycle removal ensures a DAG; layer assignment creates tiers; crossing minimization reorders nodes within layers; coordinate assignment produces final positions.

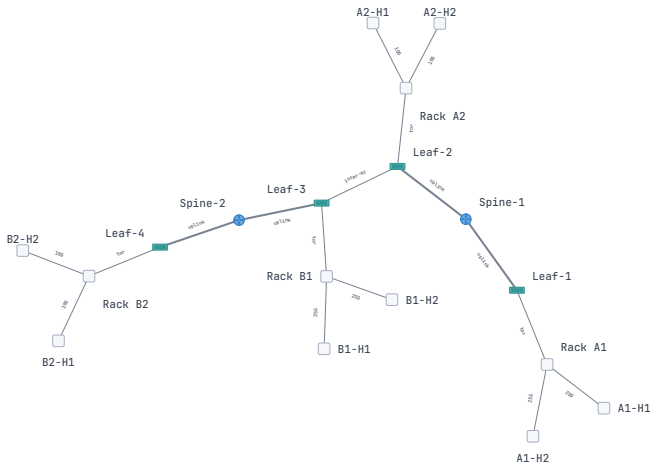


Figure 17. Hierarchical (Sugiyama) layout: a datacenter with clearly separated tiers. The algorithm minimizes edge crossings while preserving the spine → leaf → access hierarchy.

5.3 Geographic Layout

Nodes with `lat`/`lon` attributes are projected using an equirectangular projection with dynamic scale normalization $s = 1000/\max(\Delta x, \Delta y)$. Edges render as geodesic great-circle arcs (Vincenty approximation, 20 segments).

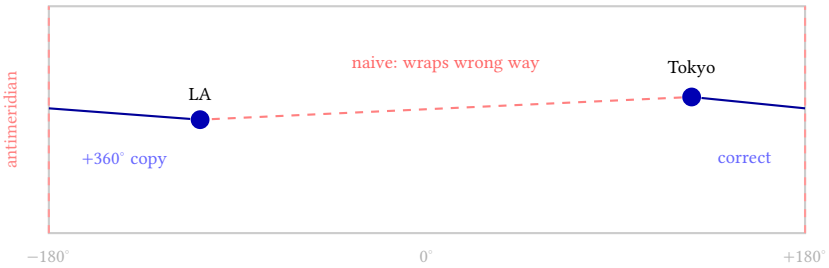


Figure 18. Antimeridian wrapping. An edge from Tokyo to Los Angeles naively crosses the full map width (red dashed). NetVis detects the projected discontinuity (>150 px jump) and renders two segments: one exiting at +180° and a 360°-shifted copy entering at −180°.

5.4 Isometric Multi-Layer Layout

Each layer runs an independent force-directed sub-layout. Results are projected isometrically with rotation angle $\phi = \pi/10$ (18°) and configurable layer spacing (default: 350 scene units). Layer planes render as filled SVG polygons. Edge labels are suppressed when edge length falls below 40 scene units.

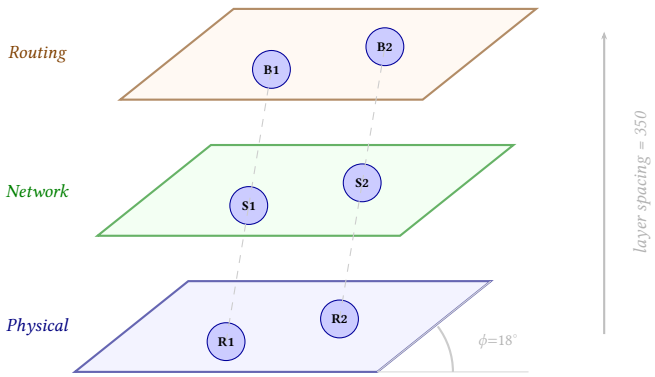


Figure 19. Isometric projection geometry. Each protocol layer runs an independent 2D layout, then all layers are projected with rotation angle $\phi = 18°$. Inter-layer edges (dashed) connect corresponding entities across planes. Layer planes render as filled SVG polygons with configurable opacity.

5.5 Orthogonal Layout

Assigns node positions to a grid and routes all edges as sequences of horizontal and vertical segments with configurable corner radius. A sweep-line algorithm minimizes total edge length while maintaining minimum channel spacing between parallel routes. Suitable for standards-compliant engineering documentation.

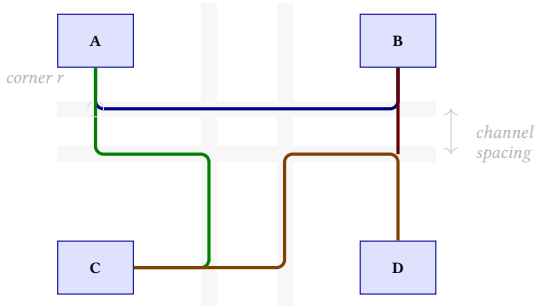


Figure 20. Orthogonal layout with channel routing. Nodes are placed on a grid; all edges are routed as horizontal/vertical segments through dedicated routing channels. A sweep-line algorithm assigns channels to minimize total edge length while enforcing minimum channel spacing. Configurable corner radius rounds the bends.

5.6 Subway Layout

Transit-map style: edges constrained to 0, 45, and 90 degrees; nodes snapped to a regular grid. The engine uses force-directed initialization followed by grid snapping, then routes edges along the octilinear grid with at-most-two-bend routing. Node positions are iteratively adjusted to minimize edge crossings while respecting the grid constraint.

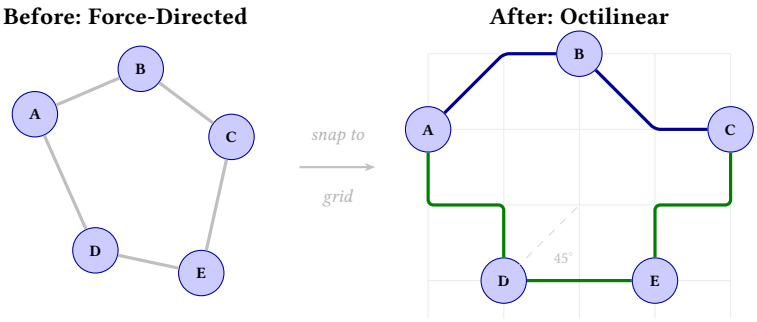


Figure 21. Subway layout: force-directed positions (left) are snapped to an octilinear grid (right). All edges use only 0°, 45°, and 90° segments with at most two bends per edge.

5.7 Radial Tree, Tree, and Composite

Radial tree places a root node at the center with children radiating outward in concentric rings; ring radii scale with subtree size to prevent overlap. **Tree layout** is a simpler algorithm for strict hierarchies: places root at top, arranges children in balanced subtree columns, and skips cycle removal. **Composite layout** assigns different layout algorithms to different sub-graphs within a single canvas, using a parent-level layout to position groups relative to each other.

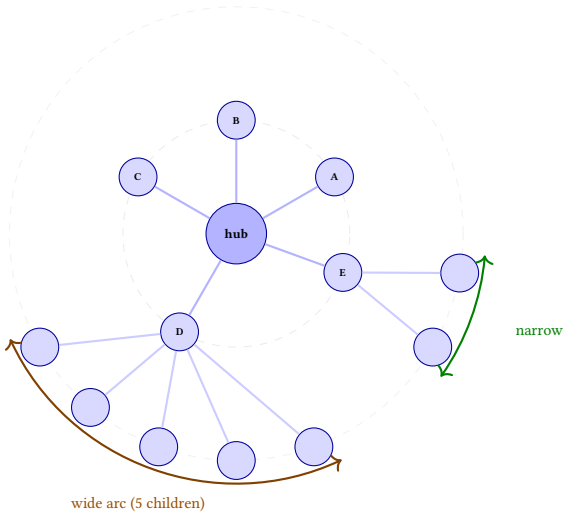


Figure 22. Radial tree layout. The hub sits at center; children radiate outward in concentric rings. Each subtree's angular allocation is proportional to its descendant count: D's 5-child subtree gets a wide arc; E's 2-child subtree gets a narrow arc. This prevents overlap without wasting space.

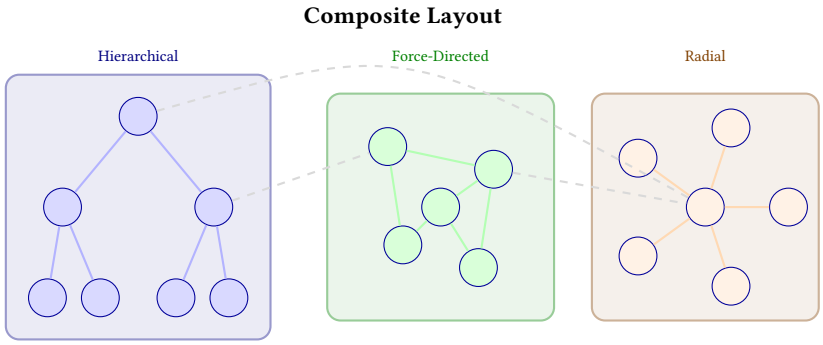
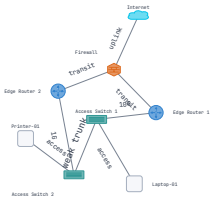


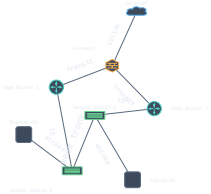
Figure 23. Composite layout: three sub-graphs use different algorithms (hierarchical, force-directed, radial) within a single canvas. A parent-level layout positions the groups relative to each other; inter-group edges (dashed) connect across boundaries.

6 Theme Gallery

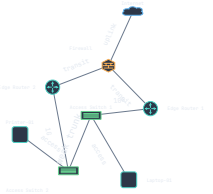
NetVis ships 7 built-in themes. Each theme defines a complete visual vocabulary: node fill and stroke colors, edge stroke and gradient defaults, label color and font family, background fill, glow and shadow intensity, and group boundary style.



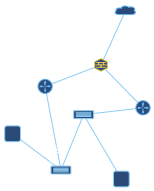
(a) light



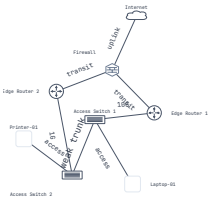
(b) dark



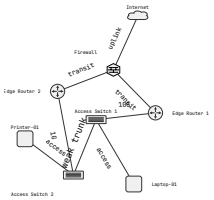
(c) network



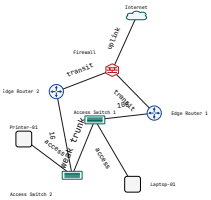
(d) blueprint



(e) datacenter



(f) print

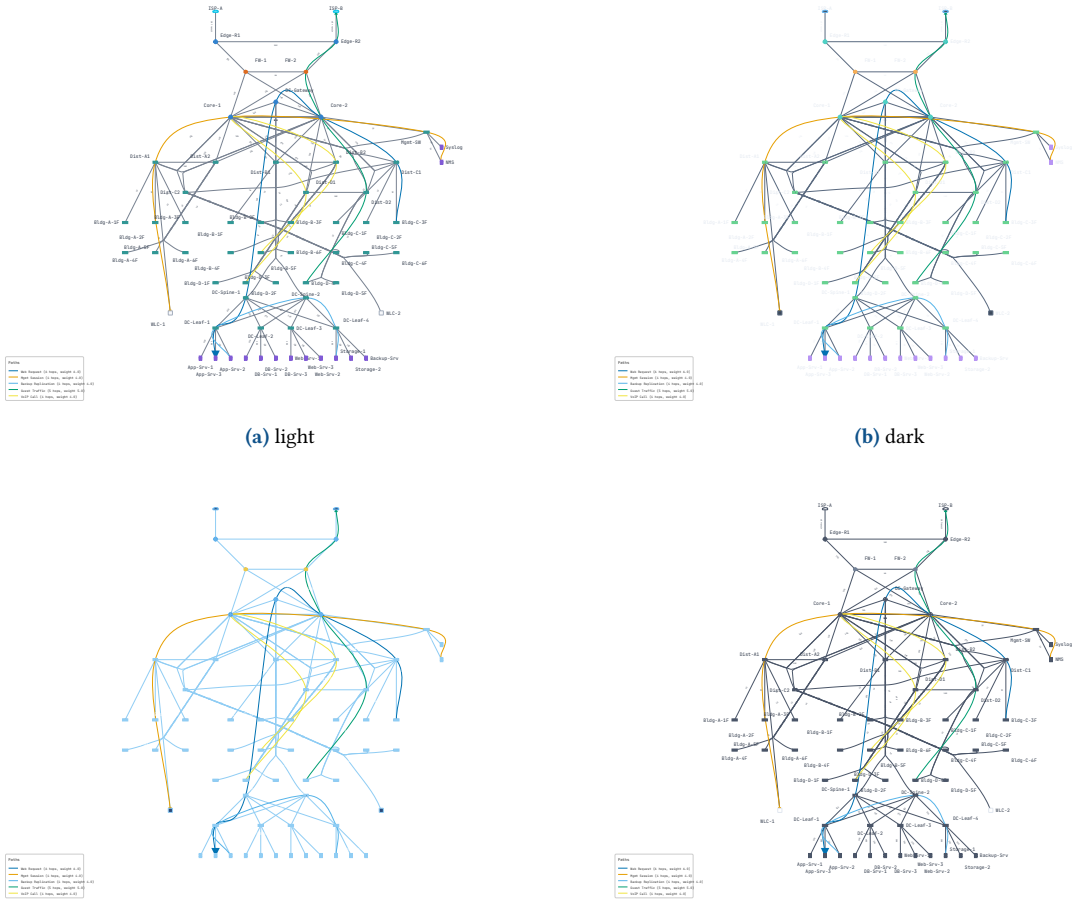


(g) high-contrast

Figure 24. All 7 built-in themes applied to the same 8-node campus topology. Theme is the only YAML difference between these diagrams. The `blueprint` theme uses a technical-ink style optimized for geometric layouts; `high-contrast` meets WCAG AA contrast ratios throughout.

6.1 Enterprise Campus Theme Comparison

Figure 25 shows the enterprise campus topology in four themes to demonstrate how the theme system adapts a complex, real-world topology.



(a) light **(b)** dark

(c) blueprint **(d)** datacenter

Figure 25. Enterprise campus (72 nodes, 98 edges) in four themes. Each theme adapts node icons, edge colors, label styling, background, glow intensity, and shadow direction. Only the `theme:` field changes in the YAML.

7 Parallel Edge Rendering

Network topologies frequently contain multiple physical links between the same pair of devices—redundant uplinks, LAG bundles, and NIC-teaming configurations. General-purpose graph tools either render these as overlapping lines (unreadable) or collapse them into a single edge (losing information).

NetVis provides five rendering modes, selectable via `parallel_mode`:

Mode	Behavior
`offset` (default)	Each link drawn as a visually separate parallel line with perpendicular offset. 4 edges produce offsets of −6, −2, +2, +6 pixels.
`bundled`	All links collapsed into one thick line (width scales with $\sqrt{\text{count}}$) with aggregate multiplicity label (e.g., “4x 10G (40G)”).
`bonding_icon`	Single line with a bracket or chain-link icon at the midpoint.
`stacked`	Lines drawn very close together (1–2 px offset), creating a thick-line effect without merging strokes.
`angled`	Curves with different angles/curvatures; distinguishes many parallel links without excessive width.

Table 2. Parallel edge rendering modes.

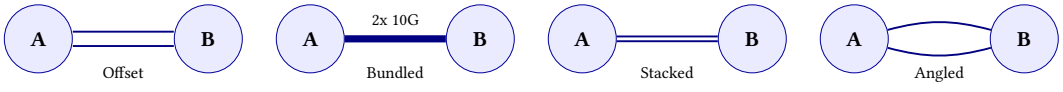


Figure 26. Four parallel edge rendering modes (BondingIcon omitted). Each mode trades visual fidelity for compactness. Offset is clearest for redundant-path diagrams; Bundled is most compact for LAG groups.

7.1 Parallel Edge Mode Decision Tree

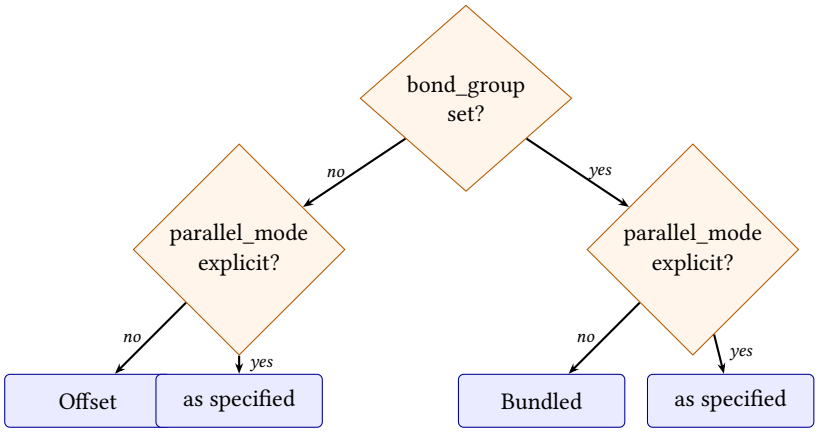


Figure 27. Parallel edge mode selection logic. Bond groups default to Bundled; non-bonded parallel edges default to Offset. An explicit `parallel_mode` always overrides the default.

7.2 Automatic Bandwidth Aggregation

When edges share a `bond_group` identifier, the system automatically parses per-link bandwidth labels and computes aggregate totals using `parse_bandwidth_gbps()`. In Bundled mode, only the total is shown (e.g., "40G"); in other bonded modes, the detail is shown (e.g., "4x 10G (40G)").

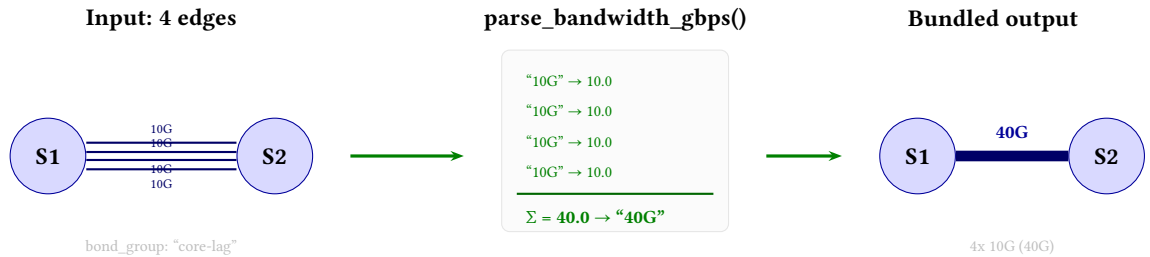


Figure 28. Automatic bandwidth aggregation. Four edges sharing `bond_group: "core-lag"` have their bandwidth labels parsed by `parse_bandwidth_gbps()`, which recognizes SI suffixes (M/G/T). The totals are summed and formatted: Bundled mode shows only "40G"; other modes show "4x 10G (40G)".

8 Node Shapes and Device Icons

NetVis supports a rich library of geometric shapes and network device icons for node rendering.

8.1 Available Shapes and Icons

Name	Category	Description
circle	Geometric	Default for generic nodes
rectangle	Geometric	Axis-aligned; useful for hosts and servers
ellipse	Geometric	Horizontal ellipse for cloud/WAN nodes
rounded_rectangle	Geometric	Rectangle with configurable corner radius
diamond	Geometric	Rotated square; used for firewalls
triangle	Geometric	Upward-pointing triangle
hexagon	Geometric	Six-sided; useful for AS/BGP speakers
router	Icon	Cylindrical icon with routing arrows
switch	Icon	Rectangular icon with port stubs
l3_switch	Icon	Switch icon with routing overlay
firewall	Icon	Shield or brick-wall icon
server	Icon	Rack-unit rectangle
cloud	Icon	Cumulus cloud outline
load_balancer	Icon	Funnel or balance icon
database	Icon	Stacked-cylinder icon
endpoint	Icon	Laptop/workstation icon
wireless_ap	Icon	Access-point radio wave icon

Table 3. Built-in node shapes and device icons. Unknown `node_type` values render as circles.

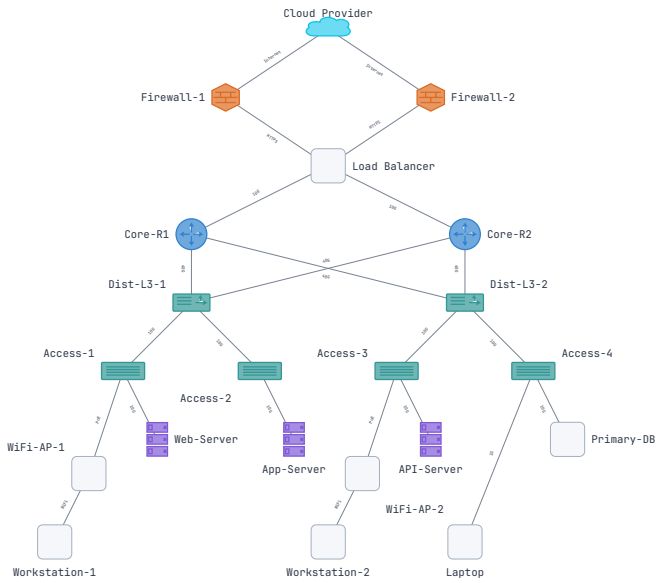


Figure 29. Full device icon library. Each icon shape adapts its size, fill, and stroke to the active theme.

9 Interface Rendering

NetVis treats network interfaces as first-class visual objects. Each node can expose its interfaces with configurable visual modes and status indicators.

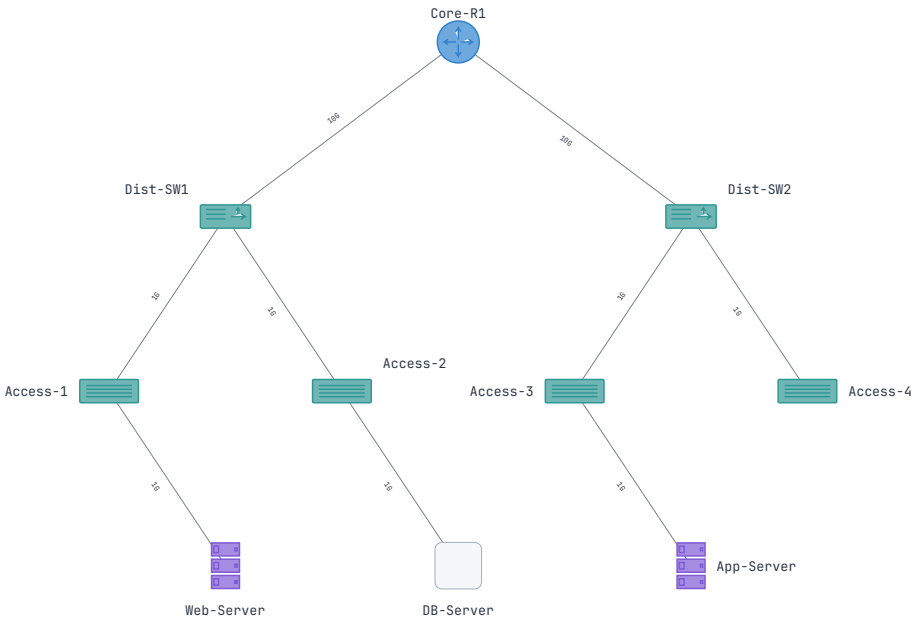


Figure 30. Interface rendering modes: Dot (compact), Port (stub with label), and Detailed (full interface card with status). Status colors are Up (active), Down (gray), AdminDown (strikethrough), and Errors (amber).

Interfaces are configured per-node in YAML:

Interface configuration

```
nodes:
  - id: core-sw
    node_type: switch
    interfaces:
      - { id: eth0, status: up,         mode: port,     marker: dot }
      - { id: eth1, status: errors,     mode: port,     marker: square }
      - { id: mgmt0, status: admin_down, mode: dot,     marker: none }
```

The `status` field accepts `up`, `down`, `admin_down`, or `errors`. The `mode` field controls visual detail: `dot` is compact (small circle on node boundary); `port` adds a rectangular stub with label; `detailed` shows the full interface card with speed and operational state. The `marker` field (`none`, `dot`, `square`, `diamond`) controls the endpoint symbol on the edge.

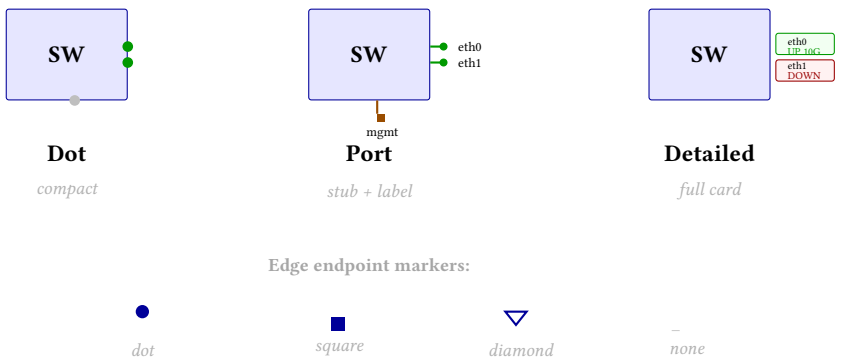


Figure 31. Interface rendering modes. **Dot:** compact circles on the node boundary. **Port:** rectangular stubs with labels. **Detailed:** full interface cards with status and speed. Four marker types control the edge endpoint symbol.

10 Edge Gradients and Path Emphasis

10.1 Edge Gradients

Directional stroke gradients encode flow direction or metric values along an edge. A gradient edge transitions smoothly from a *start color* at the source to an *end color* at the target, implemented as SVG `<linearGradient>` elements aligned to the edge direction:

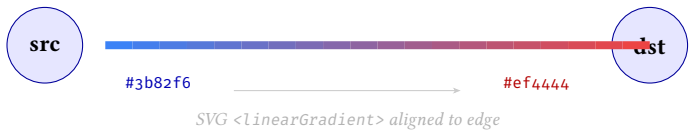


Figure 32. Edge gradient rendering. A directional gradient transitions from `start_color` at the source to `end_color` at the target, implemented as an SVG linear gradient rotated to match the edge angle.

Edge gradient

```yaml
edges:
  - from: core1
    to:   edge1
    gradient:
      start_color: "#3b82f6"  # blue at source
      end_color:   "#ef4444"  # red at target
```

10.2 Path Emphasis

Path emphasis highlights a named subset of edges as a forwarding path. Two emphasis levels (**Primary** for dominant, **Secondary** for backup) combine with three fanout modes (**Fixed**, **Scaled**, **Hybrid**) that control how emphasis interacts with parallel edge groups.

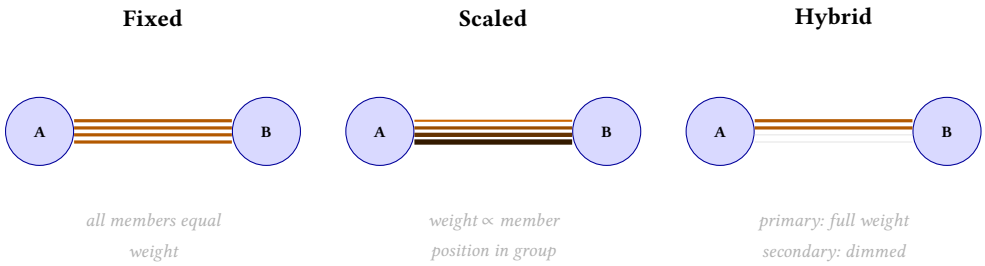


Figure 33. Path emphasis fanout modes on a 4-member LAG. **Fixed:** all members receive equal emphasis weight. **Scaled:** emphasis weight is distributed proportionally across members. **Hybrid:** primary members get full emphasis; secondary members are dimmed.

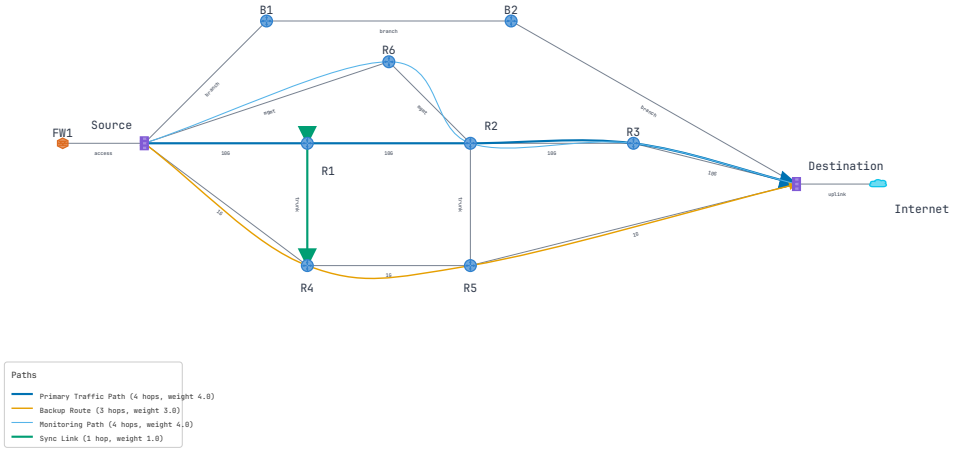


Figure 34. Path emphasis: primary paths in full-weight highlight; secondary in reduced emphasis. Fanout mode *Scaled* is shown on a LAG bundle where emphasis weight scales with member count.

11 Metric Encoding

NetVis maps arbitrary node attributes to visual channels (color, size, opacity), transforming static topology diagrams into live network dashboards. Six sequential palettes are built in: `blues`, `greens`, `oranges`, `reds`, `purples`, `greys`. All are perceptually uniform and work on both light and dark themes.

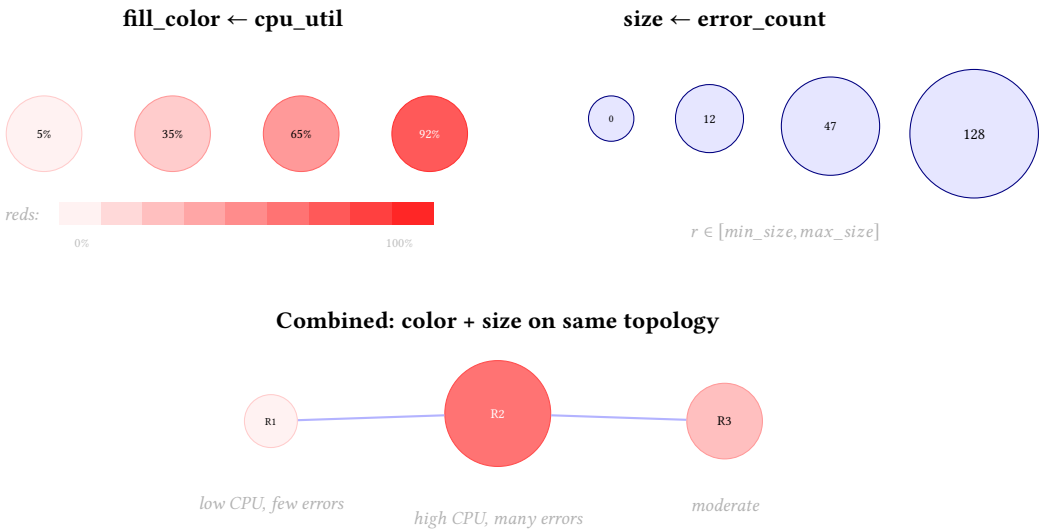


Figure 35. Metric encoding maps data to visual channels. **Top left:** CPU utilization drives node fill color through the "reds" palette. **Top right:** error count drives node radius. **Bottom:** both channels applied simultaneously—at a glance, R2 is the hotspot (large, dark red) while R1 is healthy (small, light).

Metric encoding

```
render:
  metric_encoding:
    - { attribute: cpu_utilization, channel: fill_color, palette: reds }
    - { attribute: packet_loss,     channel: size,
        min_size: 15, max_size: 40 }
    - { attribute: link_utilization, channel: edge_color, palette: oranges }
```

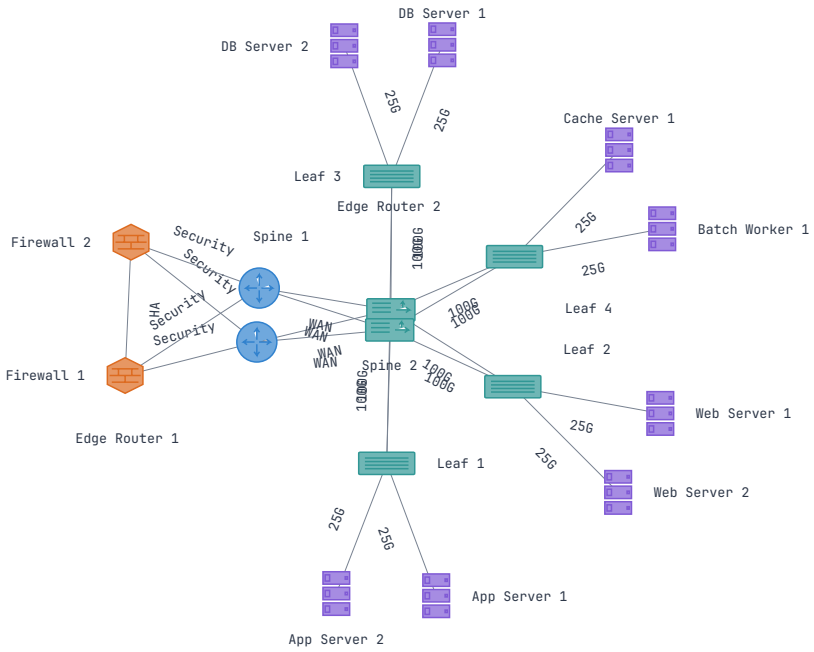


Figure 36. Metric encoding: node fill encodes CPU utilization (reds); node size encodes interface error count; edge stroke encodes link utilization (oranges).

12 Label Placement

12.1 Problem and Algorithm Choice

Placing text labels near their owner without overlapping other labels, nodes, or edges is a variant of the NP-hard point-feature label placement (PFLP) problem [11]. Three approaches were considered: (a) no collision avoidance (Graphviz approach); (b) $O(n^2)$ greedy; (c) R*-tree–accelerated greedy with $O(\log n)$ collision queries.

NetVis uses option (c). The R*-tree [12] build cost (~5 ms) is recouped after approximately 20 collision queries, which occurs by ~50 labels.

12.2 Two-Phase Greedy Algorithm

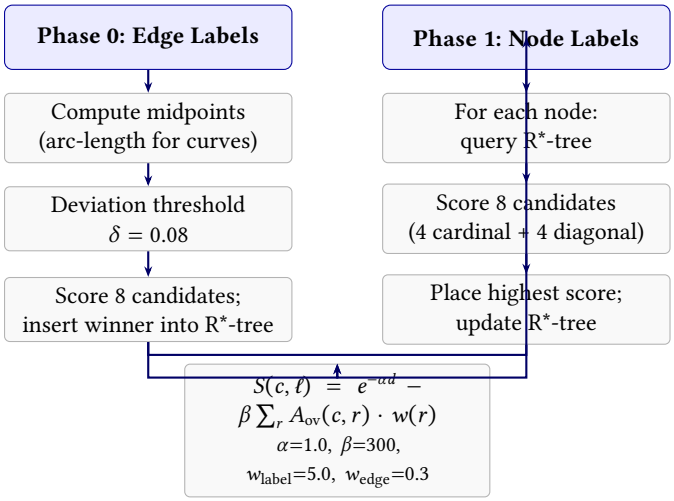


Figure 37. Label placement algorithm. Phase 0 places edge labels first (more constrained by midpoint proximity). Phase 1 places node labels around already-occupied regions. The scoring function penalizes both distance from the owner and overlap with existing geometry.

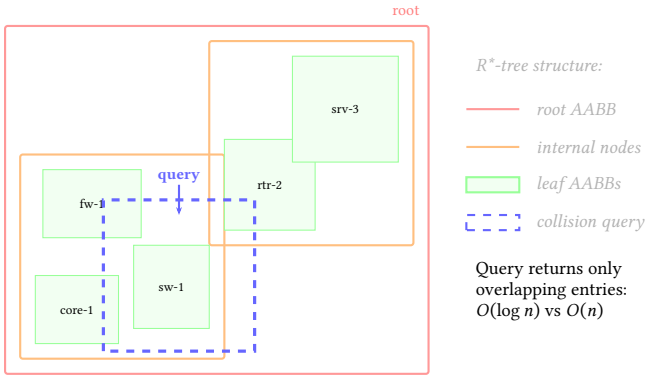


Figure 38. R*-tree spatial index for label collision detection. Occupied regions (node shapes, existing labels, edge exclusion zones) are inserted as axis-aligned bounding boxes. A candidate label position is tested by querying the tree for intersections—$O(\log n)$ per query vs. $O(n)$ brute force.

Edge exclusion zones: Curved/bundled edges can produce 47,000+ segment AABBs. Per-segment exclusion zone registration is skipped for non-straight edges; only a single bounding-box exclusion is registered per curved path.

12.3 Candidate Scoring Visualization

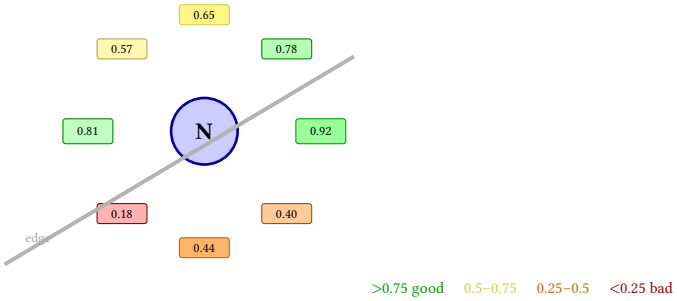


Figure 39. Scoring function applied to 8 candidate label positions around node N. Position scores (0–1) are color-coded from green (good) to red (poor). The right position wins (0.92); the lower-left loses because a diagonal edge passes through it (score 0.18).

12.4 Performance

Nodes	Time	Overlaps	Avg. Distance
100	<1 ms	0	8.2 pt
500	12 ms	0	9.1 pt
1000	48 ms	2	10.3 pt
5000	210 ms	18	12.7 pt

Table 4. Label placement performance on spine-leaf topologies (R*-tree acceleration via `rstar` [13]).

13 Edge Bundling

Dense graphs produce visual clutter from crossing edges. Edge bundling groups spatially compatible edges into bundles, reducing clutter.

13.1 FDEB Force Model

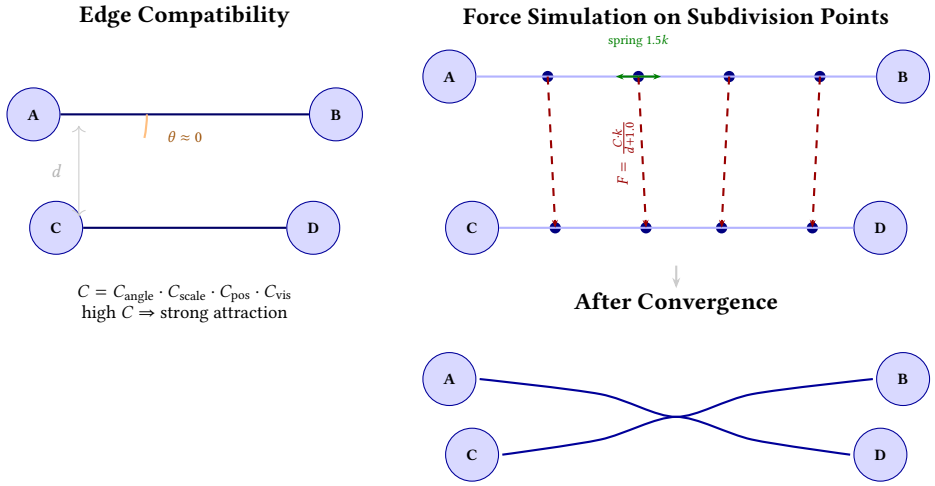
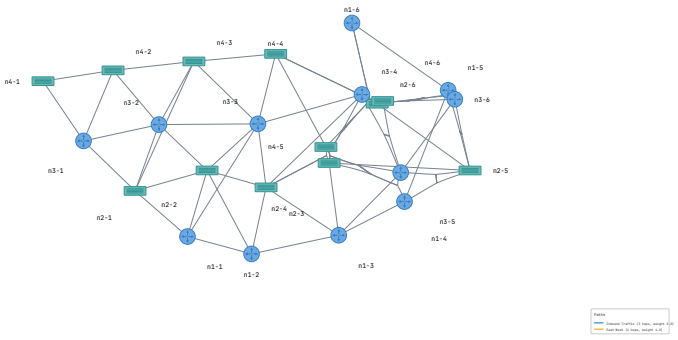


Figure 40. FDEB force-directed edge bundling. **Top left:** edge compatibility scoring—nearly parallel, co-located edges receive high compatibility C. **Top right:** subdivision points along each edge are subject to attraction forces (red, $F = Ck/(d{+}1)$) toward corresponding points on compatible edges, and spring forces (green, $1.5k$) that preserve edge shape. **Bottom:** after convergence, compatible edges merge into a shared bundle through the midpoint, fanning out to their respective endpoints.

NetVis implements a variant of Holten and Van Wijk's FDEB [14] with three key modifications: (1) singularity guard $f = k_c/(d + 1.0)$ prevents hairball artifacts; (2) spring coefficient $1.5k$ maintains edge shape; (3) Catmull-Rom tension 0.4 plus 3–5 Laplacian smoothing passes ($\alpha = 0.5$) eliminates high-frequency zigzag artifacts.

> **Note**
>
> Bundling runs *after* label placement. A `reposition_edge_labels()` pass re-places labels on edges whose curvature changed significantly post-bundling.



(a) FDEB on a spine-leaf topology

(b) Spine-leaf with shared-endpoint bundling

Figure 41. Edge bundling modes. Left: FDEB groups geometrically compatible edges. Right: shared-endpoint mode bundles edges at the hub, fanning out to spokes.

14 Edge Routing and Post-Layout Refiners

NetVis provides ten post-layout refiners, each implementing the `LayoutRefiner` trait. Refiners are composable and can be chained in any order (see §4 and Figure 14).

Refiner	Category	Description
SmartRouting	Routing	Bends edges around obstacles using R*-tree visibility graph + A* pathfinding; waypoints smoothed to Bézier splines (see Figure 42)
OrthogonalRouting	Routing	Converts edges to axis-aligned Manhattan paths with configurable corner radius
PortDocking	Routing	Adjusts endpoints to dock at specific interface positions, distributing connections evenly around the perimeter
BoundaryBundling	Routing	Routes edges through a shared crossing point at group boundaries, fanning out inside
Deoverlap	Spacing	Pushes overlapping nodes apart using repulsive spring force; configurable margin (default: 10 units)
EdgeClipping	Spacing	Clips edge endpoints to the exact node shape boundary for correct arrowhead placement
Nudging	Spacing	Nudges overlapping or near-overlapping edges apart; used after parallel edge grouping
SmartLabel	Labels	8-position snapping; second-pass improvement over initial greedy placement
LabelNudging	Labels	Iterative repulsive force between overlapping label bounding boxes; converges in 10–30 iterations
Starburst	Groups	Positions nodes radially within containment groups; best for AS-level diagrams

Table 5. Post-layout refiners.

14.1 Smart Routing: Visibility Graph

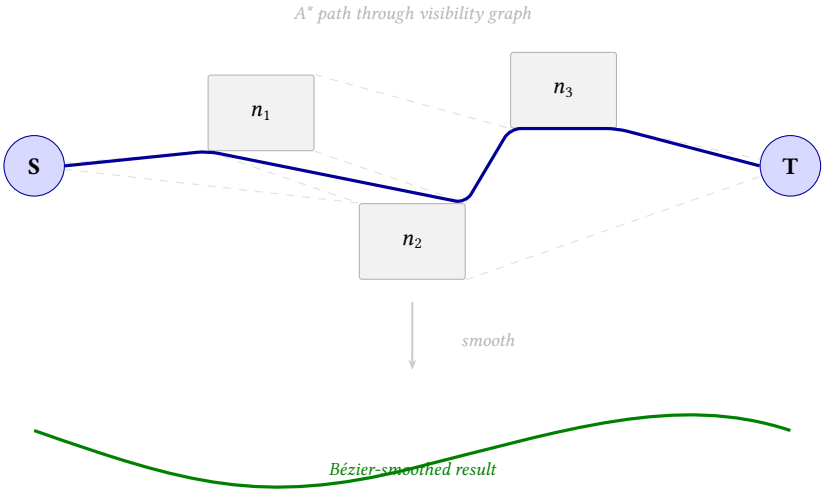


Figure 42. SmartRoutingRefiner. An R*-tree query identifies obstacles near each edge. A visibility graph connects obstacle corners; A* finds the shortest obstacle-free path. Waypoints are smoothed to Bézier splines for natural-looking curves.

14.2 Edge Clipping: Before and After

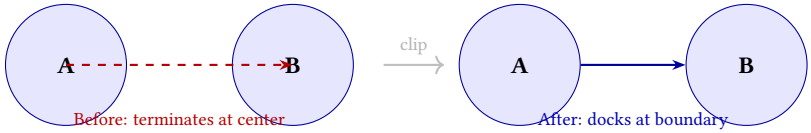


Figure 43. EdgeClippingRefiner: without clipping, edge paths terminate at node centers; with clipping, endpoints dock at the exact node shape boundary, giving correct arrowhead placement regardless of shape.

14.3 Deoverlap: Before and After

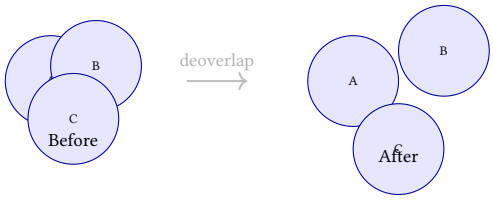


Figure 44. DeoverlapRefiner: applies a repulsive spring force between overlapping node bounding boxes, pushing them apart until the configured minimum margin is satisfied.

14.4 Boundary Bundling: Group Edge Routing

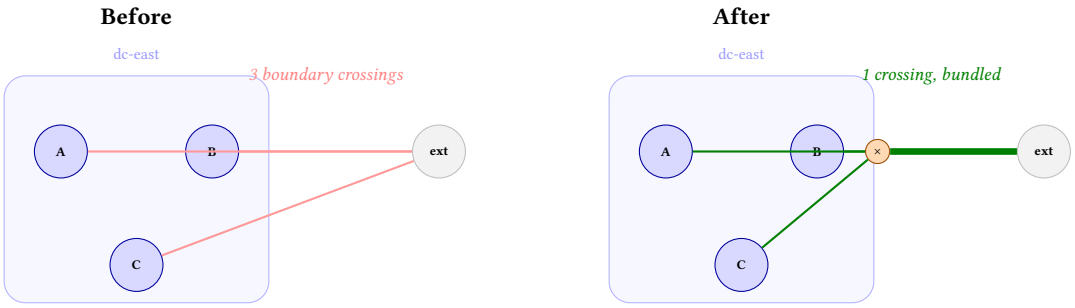


Figure 45. BoundaryBundlingRefiner. **Before:** three edges from internal nodes to an external node each cross the group boundary independently. **After:** all edges fan into a shared crossing point on the boundary and exit as a single bundled edge, reducing visual clutter at group borders.

15 Topology Diffing

The `diff` module computes added, removed, and modified nodes and edges between two `NetVisGraph` instances (by ID and attribute comparison). The `apply_diff_styles()` function color-codes the rendered scene: green for added, red/dimmed for removed, amber for modified. A diff legend is automatically inserted.

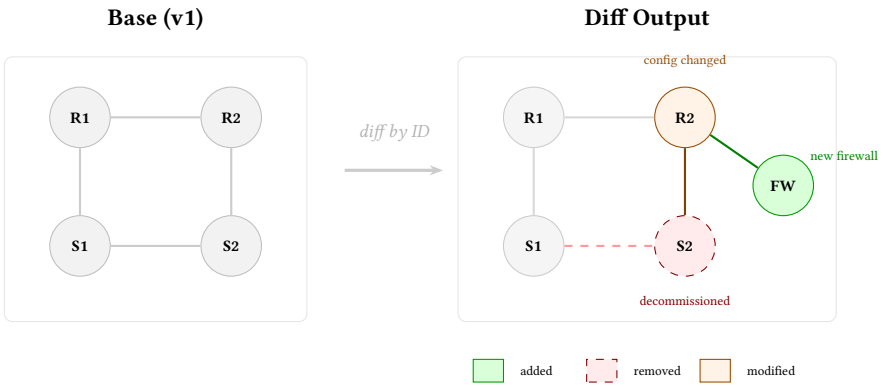


Figure 46. Topology diffing by example. The base topology (v1) is compared to a new version by matching node and edge IDs. The diff output color-codes changes: S2 was decommissioned (red/dashed), R2's config changed (amber), and a new firewall was added (green). Unchanged elements remain gray.

CLI topology diff

```
netvis diff old.yaml new.yaml -o diff.svg
```

16 Temporal Analysis

The `timeline` module tracks topology evolution over time. `discover_snapshots()` scans a directory for YAML topology files, extracts timestamps from filenames or metadata, and returns them chronologically. `build_node_history()` and `build_edge_history()` trace an entity's lifecycle: first appearance, all attribute changes, and removal.

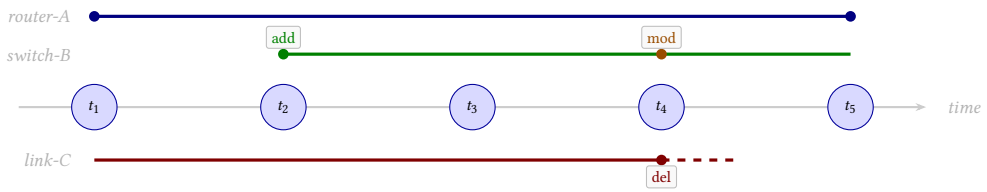


Figure 47. Temporal analysis: entity lifecycle tracking across topology snapshots. Each snapshot (t_1–t_5) is a YAML file. The timeline module traces first appearance (add), attribute changes (mod), and removal (del) for every node and edge.

Timeline CLI

```
netvis timeline --dir snapshots/
netvis query --dir snapshots/ --node router-fra-01
```

17 Traffic Flow Animation

The `traffic` module injects animated flow overlays into rendered SVGs via CSS `@keyframes`. Each edge can carry utilization (0.0–1.0), direction (`forward`/`reverse`/`bidirectional`), and animation style (`dot`/`dash`). Utilization maps to color: green (low) → amber → red (congested). Zero-traffic edges receive CSS dimming (`opacity: 0.3`). Play/pause and speed controls are injected as an SVG overlay.

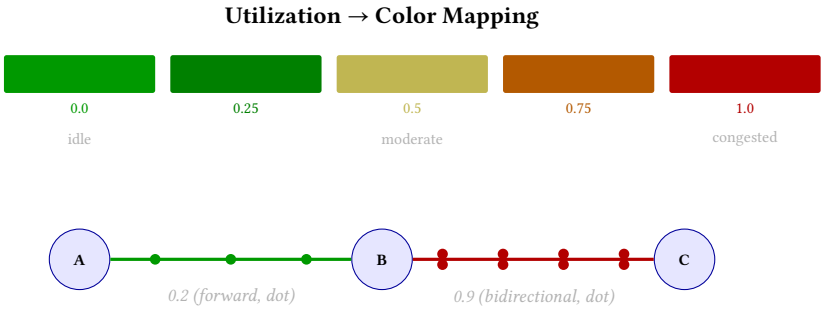


Figure 48. Traffic flow animation. Edge utilization (0.0–1.0) maps to a green–amber–red color ramp. Animated dots travel along the edge path; bidirectional mode shows two opposing dot streams. CSS `@keyframes` drives the animation in SVG output.

Traffic overlay

```
traffic:
  edges:
    - { id: core-to-agg, utilization: 0.75,
        direction: bidirectional, style: dot }
```

18 Annotation Markup

The `annotation` module injects text callouts into rendered SVGs, with pointer lines from callout to anchor node. Annotations support layer filtering (`show_only_layers`, `hidden_layers`) and three shapes: `circle`, `box`, `arrow`.

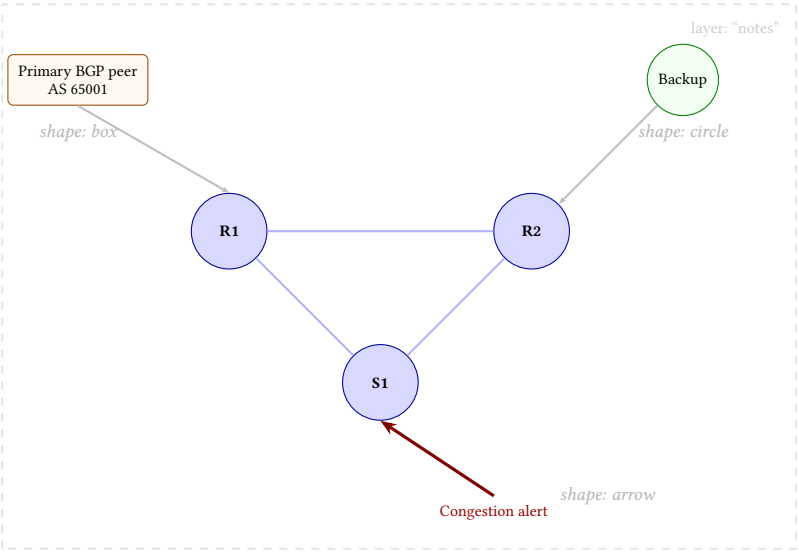


Figure 49. Annotation callout shapes. Each annotation anchors to a node with a pointer line. Three shapes are supported: **box** (rectangular, for multi-line text), **circle** (compact labels), and **arrow** (directional emphasis). Annotations are organized into named layers for selective visibility.

Annotation configuration

```
annotations:
  - anchor: router-fra-01
    type: callout
    text: "Primary BGP peer"
    shape: box
    layer: notes
```

19 Interactive Editor

The `editor` module provides interactive topology editing for browser deployments via the WASM target. It implements the command pattern: `MoveNodeCommand`, `TransactionCommand`, etc., storing only deltas (24 bytes for a node move) rather than full snapshots. An `UndoStack` with 50-action capacity provides undo/redo, using ~41,000x less memory than snapshot-based undo at 5,000 nodes.

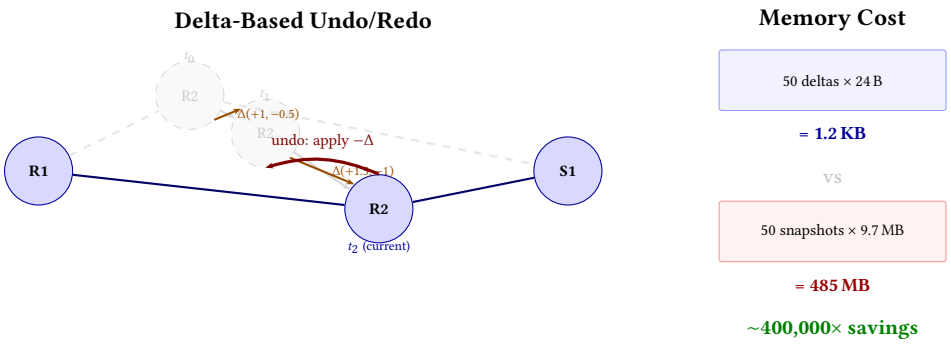


Figure 50. Interactive editor with spatial undo. Node R2 is dragged through three positions; each move stores only a delta ($\Delta x, \Delta y$, 24 bytes) rather than a full scene snapshot. Undo applies $-\Delta$ to restore the previous position. At 5,000 nodes the delta approach uses 1.2 KB vs. 485 MB for snapshot-based undo.

A Lit-based `<netvis-editor>` Web Component wraps the WASM editor:

Web Component

```
<netvis-editor topology="campus.yaml"
               layout-algorithm="force-directed">
</netvis-editor>
```

20 Visual Effects

NetVis provides a pluggable visual effects system mapping semantic effect names to SVG filter definitions. The `EffectRegistry` deduplicates identical filter definitions via content-addressed hashing. Effects are assigned via the `highlight` field:

Assigning effects

```yaml
nodes:
  - id: core-router
    highlight: true              # default glow
  - id: alert-node
    highlight: "emphasis"        # glow-emphasis
  - id: combo-node
    highlight: "stack:drop-shadow-subtle+glow-selection"
```

Name	Type	Description
`drop-shadow-subtle`	Drop shadow	Light offset shadow
`drop-shadow-emphasis`	Drop shadow	Strong offset shadow
`inner-shadow-boundary`	Inner shadow	Inset shadow for group boundaries
`glow-selection`	Glow	Selection highlight
`glow-status`	Glow	Status indicator
`glow-emphasis`	Glow	Emphasis marker
`stack:<a>+<b>`	Composed	Combine any two base effects

Table 6. Built-in visual effects.

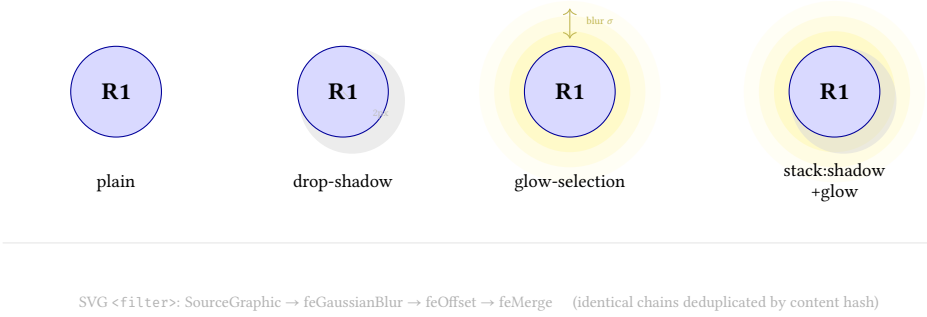


Figure 51. Visual effects applied to the same node. **Plain:** no filter. **Drop shadow:** offset blurred copy behind the node. **Glow:** colored Gaussian blur halo radiating outward. **Composed:** `stack:` combines both via SVG `feMerge`. The EffectRegistry deduplicates identical filter chains by content hash.

All effects adapt to the active theme. Glow intensity and shadow direction vary by theme family.

Effect Budget

SVG filter performance degrades at ~50+ filtered elements. `effects.max_filtered_elements` (default: 50) sets a budget.

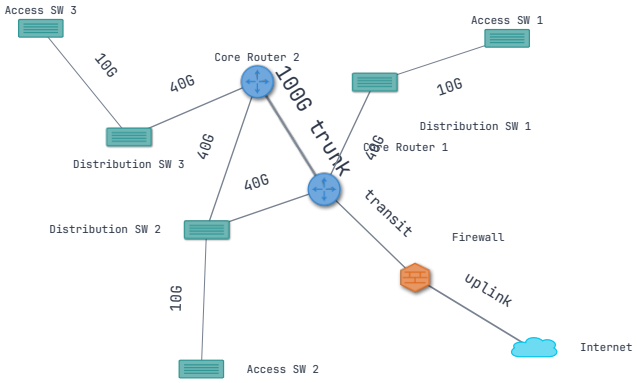


Figure 52. Visual effects showcase. Left to right: drop shadow, glow selection, emphasis glow, and composed `stack:drop-shadow+glow`. All effects adapt their intensity and color to the active theme.

21 Graph Analytics

The `analytics` module computes structural graph metrics and injects them as a visual "Network Insights" card. Three metrics are computed: **betweenness centrality** (Brandes' algorithm, $O(nm)$) identifies critical transit nodes; **graph diameter** (longest shortest path) highlights worst-case latency; **clustering coefficient** measures local redundancy.

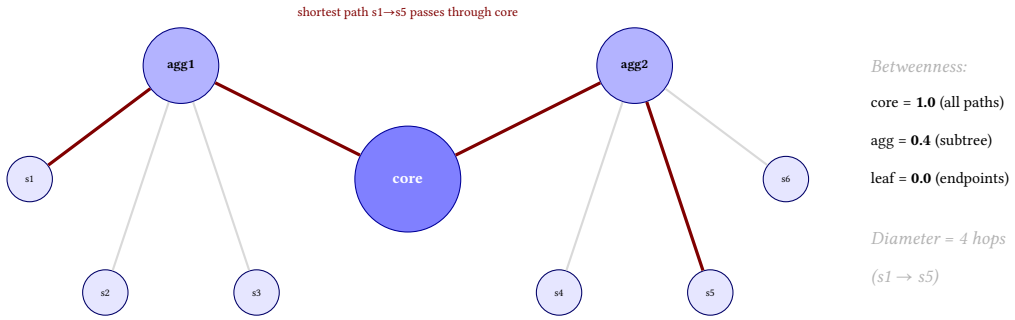


Figure 53. Graph analytics: betweenness centrality visualized. Node size encodes centrality—the core router sits on every shortest path between the two subtrees (centrality 1.0), making it the critical transit point. Aggregation nodes score 0.4 (bridging their subtrees); leaf nodes score 0.0. Graph diameter (longest shortest path) is 4 hops.

Analytics CLI

```
netvis --input topology.yaml --analytics -o analytics.svg
```

22 Filter DSL

NetVis includes a Domain-Specific Language (DSL) for filtering topology elements before rendering. The system comprises three layers: a **parser** that converts filter expressions into an AST (supporting boolean operators, attribute selectors, glob patterns, and set operations); an **evaluator** that walks the AST per node/edge; and an optional **LLM layer** (feature-gated) that translates English queries into filter AST nodes.

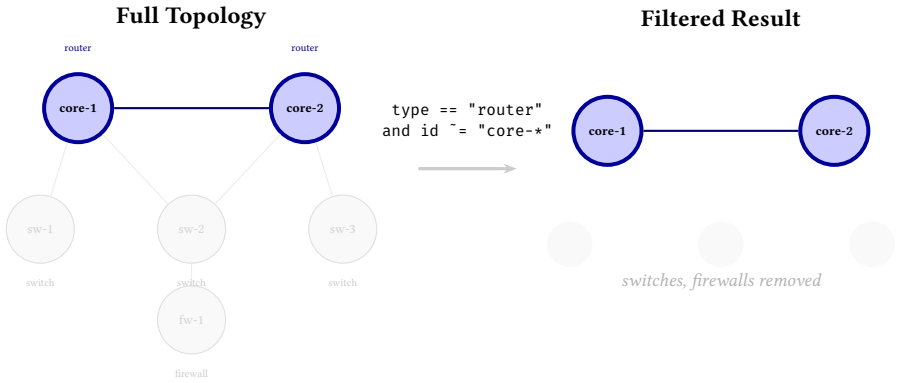


Figure 54. Filter DSL in action. The expression `type == "router" and id ~= "core-*"` matches only the two core routers (highlighted). All non-matching nodes and their incident edges are removed from the rendered output. The filter supports boolean operators, attribute comparisons, and glob patterns.

Filter DSL examples

```
netvis --input t.yaml --filter 'type == "router"'
netvis --input t.yaml \
  --filter 'group == "dc-east" and id ~= "core-*"'
```

23 Input Validation

Topology and configuration files are validated against a JSON Schema (generated from Rust types via `schemars`). The `validation::fuzzy` module provides "Did you mean…?" suggestions for misspelled YAML field names using Levenshtein distance (`strsim` crate). Structured diagnostics are returned as `ValidationDiagnostic` entries with severity, location, and suggested corrections.

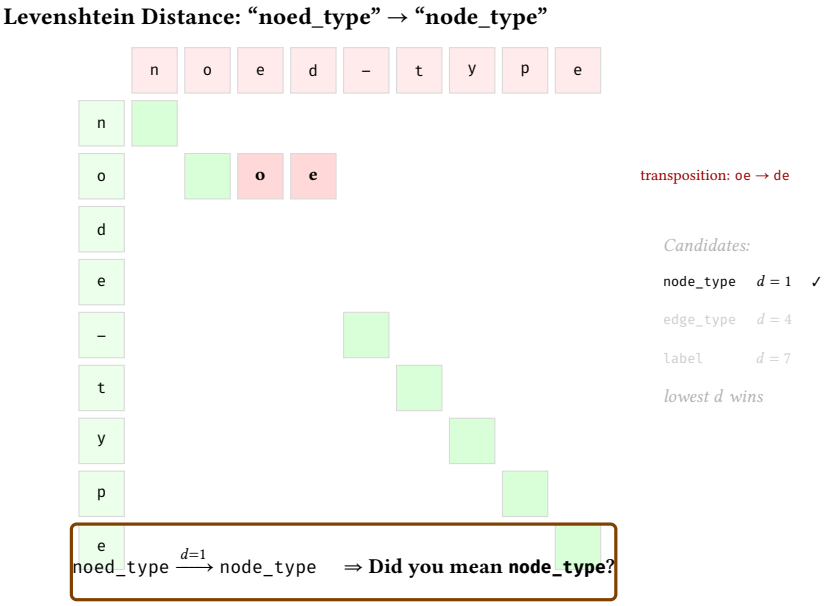


Figure 55. Fuzzy validation with Levenshtein distance. The misspelled field `noed_type` is compared against all valid schema fields. Character-level alignment reveals a single transposition (oe→de), yielding distance $d = 1$. The closest match (`node_type`) is suggested as a correction.

Schema export and validation

```
netvis schema --type topology > topology-schema.json
netvis validate --input topology.yaml
```

24 Quality Metrics

NetVis computes quality metrics as part of every render, following the aesthetic metrics framework of Purchase [15] and Dunne et al. [16].

24.1 Metric Families

Metric	Description
Edge crossings	R*-tree segment intersection, weighted by angle
Overlap area	AABB intersection between all label/node pairs; sub-pixel (<2 px) ignored
Label readability	Composite of owner distance, font-size variance, reading order
Angular resolution	Minimum angle between consecutive incident edges
Node alignment	Degree to which nodes align on axes or grid lines
Edge clearance	Minimum distance from non-incident edges to nodes
Coincident edges	Detection of duplicate or overlapping edge paths
Contrast	Label color contrast against background (WCAG-derived)
Edge-length variance	Coefficient of variation across edge lengths
Spatial distribution	Evenness of node distribution (entropy-based)
Path straightness	Curvature penalty for bent edges
Viewport coverage	Elements outside viewport bounds
Node–edge overlap	Non-incident edge segments crossing node bounding boxes
Density heatmap	Spatial density for identifying visual hotspots

Table 7. 14 of the 16 quality metric families computed per render. All are aggregated into a unified `AestheticScore` (0–100).

24.2 Clarity Score and CI Integration

The composite *Clarity Score* (0–100) combines four equally-weighted factors: edge crossings, angular resolution, node spacing, and label overlap. It is designed as a single-number summary for CI/CD pipelines.

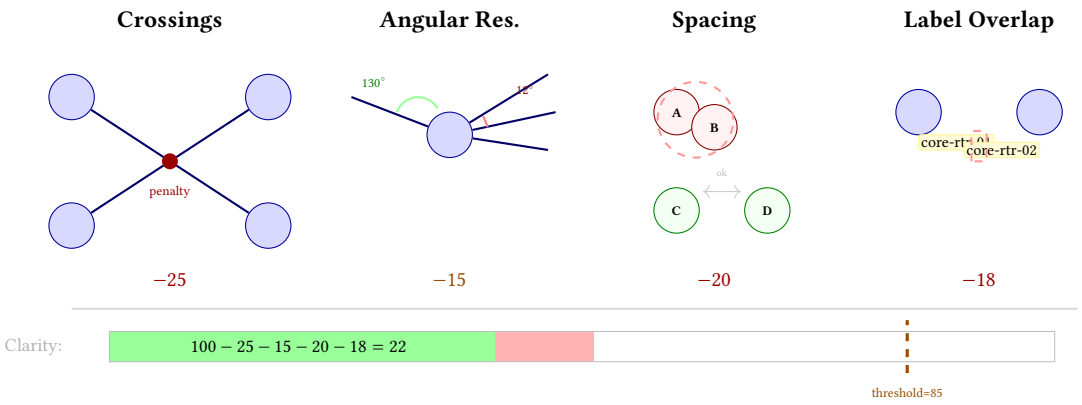


Figure 56. Clarity score: four quality factors illustrated spatially. Each panel shows the visual defect and its penalty. Edge crossings, narrow angular resolution, overlapping nodes, and colliding labels all subtract from 100. CI/CD pipelines gate on a configurable threshold (default: 85; this example scores 22—a clear failure).

Quality baselines are stored in `benches/regression_baselines.json`; CI fails if any metric degrades by more than 20%.

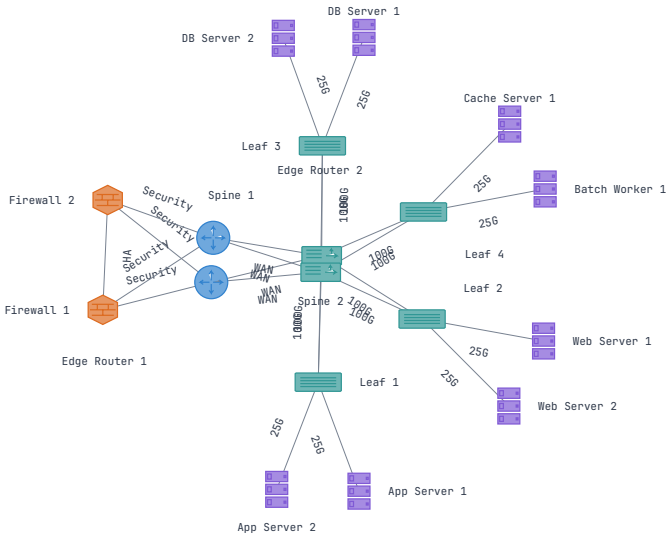


Figure 57. Quality metrics are computed alongside every render and can be overlaid on the output. Each node's betweenness centrality, edge utilization, and label overlap are directly visible.

Separate from aesthetic quality, the `lint` module performs structural topology checks: isolated nodes, disconnected subgraphs, duplicate edge IDs, and missing labels. Lint results appear as `Warning`-severity entries in the `DiagnosticsEnvelope`.

25 Rendering Pipeline

25.1 SVG Renderer

The primary renderer produces SVG using the `svg` crate. Elements render in painter's-model order: background map layers → isometric layer planes → group boundaries → edges → nodes → labels → effects.

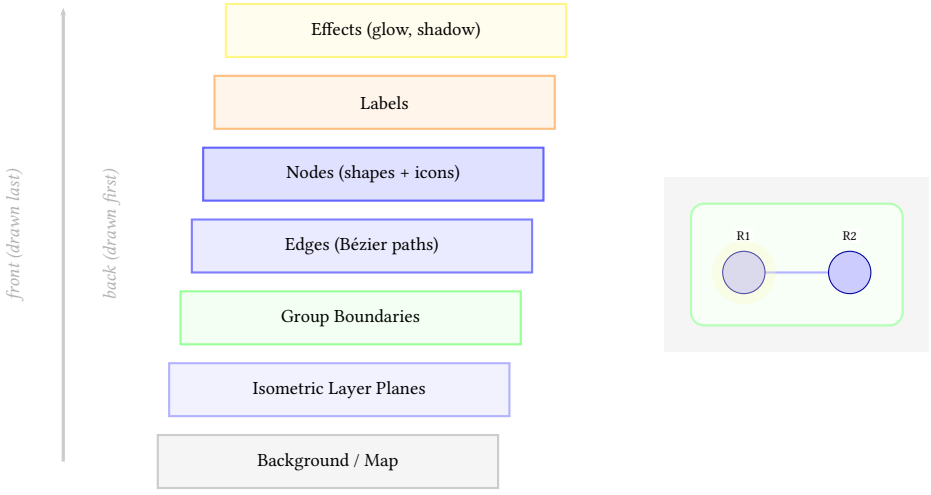


Figure 58. SVG rendering layer order (painter's model). Elements are drawn back-to-front: background first, effects last. This ensures labels are always readable over edges, nodes always cover edge strokes at endpoints, and effect overlays (glow, shadow) render above all content.

25.2 Level of Detail

LOD suppresses visual details at configurable thresholds, following Abello et al. [17]:

Profile	Labels	Edge Lbl.	Interfaces	Effects
Conservative	2000	1000	1000	500
Balanced	500	300	300	200
Aggressive	200	100	100	50

Table 8. LOD suppression thresholds.

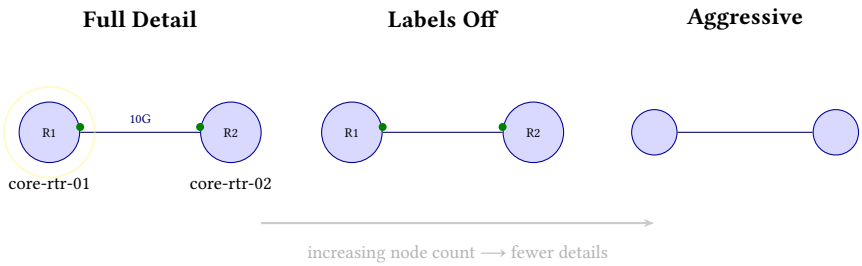


Figure 59. Level of detail (LOD) progressive suppression. As topology size grows past profile thresholds, labels, edge labels, interface markers, and visual effects are progressively hidden to maintain render performance and visual clarity.

25.3 Terminal Renderer

`netvis show` renders topology to Unicode box-drawing and ANSI color codes without a graphical display. `light` produces plain ASCII; `dark` produces ANSI-colored output.

25.4 Export Formats

SVG is the canonical output. Additional formats: PNG via `resvg`; PDF via `svg2pdf`; self-contained interactive HTML (WASM embedded as base64, pan/zoom/drag, openable from `file://`); Graphviz DOT for interoperability.

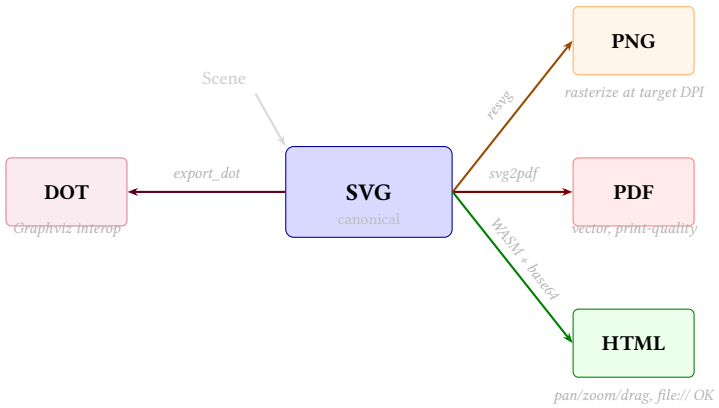


Figure 60. Export format pipeline. SVG is the canonical intermediate representation. `resvg` rasterizes to PNG at configurable DPI; `svg2pdf` produces vector PDF for print; the HTML exporter embeds WASM as base64 for interactive pan/zoom/drag openable from `file://`; DOT export enables Graphviz interoperability.

26 Pluggable Node Renderer

The `NodeRenderer` trait defines the interface for custom node renderers. Built-in types (router, switch, server, firewall, cloud, l3_switch, and the full icon library from §8) are registered at startup. Custom renderers are registered with the `NodeRendererRegistry`; unknown `node_type` values fall back to the default circle renderer.

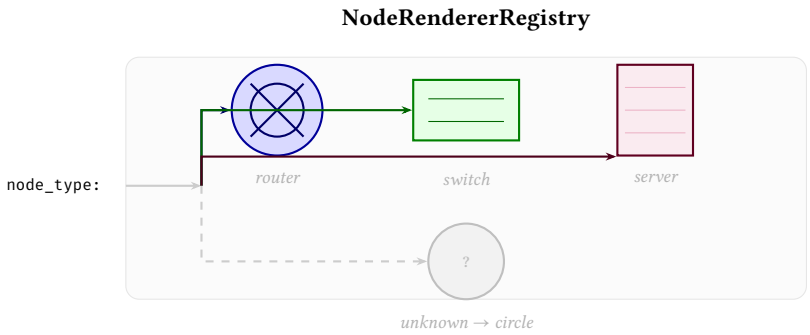


Figure 61. Pluggable node renderer dispatch. The `NodeRendererRegistry` maps `node_type` strings to shape renderers. Built-in types (router, switch, server, firewall, cloud, l3_switch) are registered at startup. Unknown types fall back to the default circle renderer, ensuring graceful degradation for unrecognized device types.

27 API and Usage

27.1 Rust API

Function	Input	Returns
`nv::render`	Graph, Options	RenderEnvelope
`Layout::layout`	Graph	Scene
`SvgRenderer::render`	Scene	SVG string
`render_to_png`	SVG, PngOptions	PNG bytes
`render_to_pdf`	SVG, PdfOptions	PDF bytes
`export_dot`	Scene	DOT string
`ExportBuilder::build`	Config	Multi-format output

Table 9. Core Rust API entry points.

27.2 WASM / JavaScript

JavaScript integration

```
import { init, renderToSVG, renderWithDiagnostics } from '@netvis/core';
await init();
const { output, diagnostics } = await renderWithDiagnostics(topologyJson);
```

A background Web Worker API (`layoutWorkerRun`) offloads layout computation with progress callbacks streaming intermediate node positions. See Appendix B for the full CLI reference.

28 Deployment

NetVis compiles from a single Rust codebase to three targets:

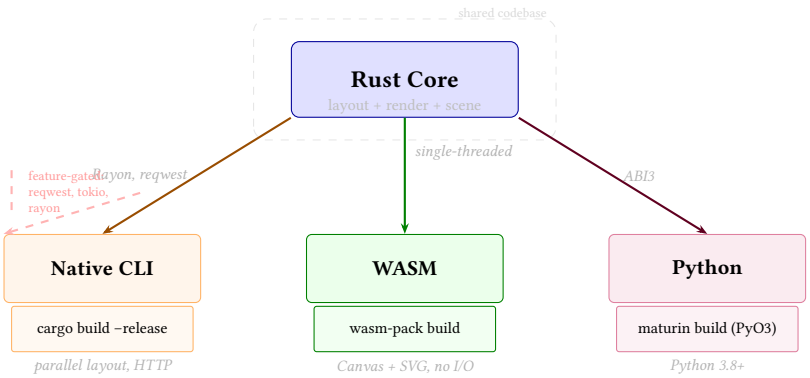
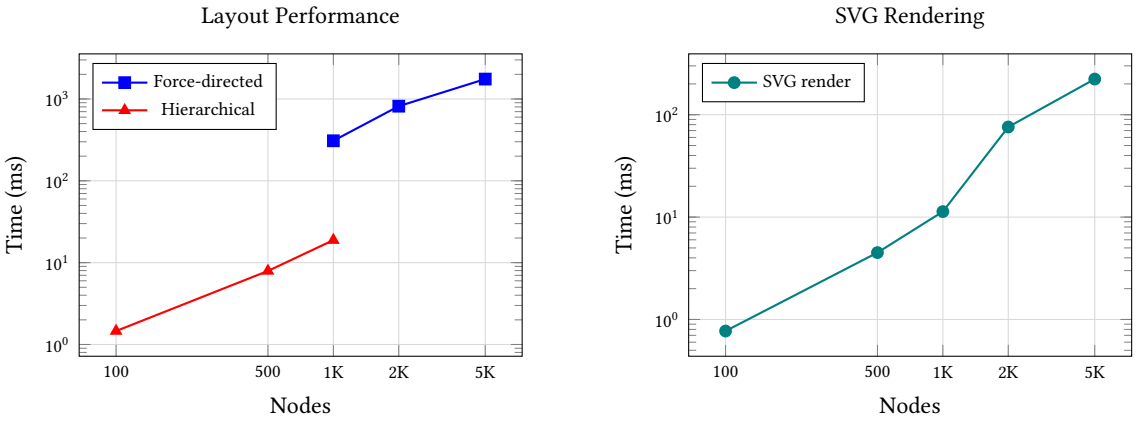


Figure 62. Deployment architecture. A single Rust codebase compiles to three targets via feature flags. WASM-incompatible crates (reqwest, tokio, rayon) are gated to native-only. The Python binding uses PyO3 with ABI3 stability for broad version support.



(a) Layout algorithm scaling.

(b) SVG rendering scaling.

Figure 63. NetVis benchmark results on spine-leaf topologies. Left: layout computation time for force-directed (Barnes–Hut) and hierarchical (Sugiyama) engines. Right: SVG rendering time including label placement. Both axes are log-scaled; data points are median wall-clock times.

Target	Build command	Notes
Native CLI	`cargo build --release --features cli`	Full feature set; parallel layout via Rayon; HTTP adapters
WASM [18]	`wasm-pack build --target web`	Single-threaded; Canvas + SVG rendering; no file I/O
Python	`maturin build --features python`	PyO3, ABI3 stability (Python 3.8+)

Table 10. Deployment targets and build commands.

WASM-incompatible crates (`reqwest`, `tokio`, `rayon`) are gated to native-only via feature flags. Deterministic mode (CLI: `--deterministic`; YAML: `render.seed`) ensures reproducible output for CI/CD pipelines.

29 Performance

Measured on Apple M4 Pro, 24 GB RAM, macOS, Rust 1.93.

Operation	100	500	1K	2K	5K
Force-directed layout	20 ms	144 ms	309 ms	816 ms	1.75 s
Hierarchical layout	1.5 ms	8 ms	19 ms	—	—
SVG render	0.8 ms	4.5 ms	11 ms	76 ms	223 ms
Peak heap allocation	215 KB	952 KB	1.9 MB	3.8 MB	9.7 MB

Table 11. Performance baselines by topology size (spine-leaf topologies).

10,000-node spine-leaf (Enterprise engine, auto LOD): 5.3 s end-to-end.

Tool	100 nodes	1K nodes	2K nodes
Graphviz `sfdp`	84 ms	250 ms	914 ms
Graphviz `neato`	88 ms	1,024 ms	4,434 ms
NetVis (full pipeline)	**24 ms**	**330 ms**	**620 ms**

Table 12. Comparative benchmark on identical spine-leaf topologies. NetVis times include layout, label placement, styling, and SVG rendering.

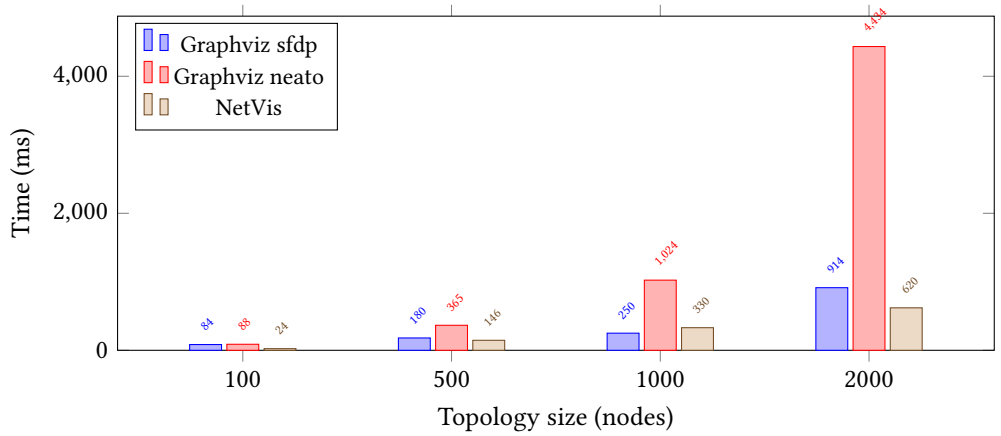


Figure 64. Comparative performance on identical spine-leaf topologies. NetVis includes the full pipeline (layout, labels, styling, SVG render); Graphviz includes layout and SVG output only.

30 Conclusion

This report has presented NetVis, a 90,000+ line Rust-based network topology visualization engine. Four contributions distinguish it from prior work:

1. **Dual-engine force-directed layout.** A Standard engine for topologies up to 2,000 nodes and a Barnes–Hut-accelerated Enterprise engine scaling to 10,000+ nodes (§5), complemented by six additional layout algorithms (hierarchical, geographic, isometric, orthogonal, subway, radial/tree).
2. **R*-tree–accelerated label placement.** A two-phase greedy algorithm using spatial indexing for $O(\log n)$ collision queries, placing labels for 5,000-node topologies in under 200 ms (§12).
3. **Composable refiner pipeline.** Ten post-layout refiners—label placement, edge bundling, smart routing, deoverlap, nudging, boundary bundling, and others—that compose in user-specified order to separate layout concerns from visual polish (§4, §14).
4. **Automated quality metrics.** Sixteen metric families (crossing count, angular resolution, overlap ratio, label clarity, and others) computed in-pipeline, enabling regression testing and objective comparison (§24).

Benchmarks on identical spine-leaf topologies show NetVis 1.5–7× faster than Graphviz sfdp/neato while producing the full rendering pipeline (layout, labels, styling, SVG output) rather than layout alone (§29). The system compiles from a single codebase to native CLI, WebAssembly, and Python targets (§28), and exposes domain-specific features—parallel edge rendering with automatic LAG aggregation (§7), interface rendering (§9), topology diffing (§15), traffic animation (§17), and metric encoding (§11)—that general-purpose graph tools lack.

31 Limitations

Limitation	Description	Workaround / Plan
Scale (>10K nodes)	Enterprise engine handles 10K nodes in ~5 s; 50K+ would benefit from GPU acceleration	Use hierarchical/geographic/tree layout; GPU via WebGPU is under investigation
Greedy label placement	Sequential placement; pathological dense clusters may produce overlaps	Suppress labels via LOD; increase canvas size; simulated-annealing post-pass planned
Single-threaded WASM	Rayon not available in browser; large layouts slower	Use Web Workers for background computation; WASM threads pending
Non-determinism	HashMap iteration order is non-deterministic by default	Use `--deterministic` or `render.seed`
Adapter coverage	Only YAML, NetBox, LLDP/CDP built in	`Adapter` trait is public; community adapters encouraged
No user study	Quality claims based on automated metrics only	Controlled user study planned

Table 13. Known limitations and workarounds.

32 Future Work

1. **GPU-accelerated layout.** WebGPU compute shaders for 50,000+ node topologies.
2. **Streaming topology ingestion.** Live gNMI/gRPC streaming extending the timeline module (§16) to real-time updates.
3. **Simulated-annealing label post-pass.** Global optimization resolving remaining overlaps in dense clusters.
4. **User study.** Controlled perceptual evaluation with network engineers.
5. **Collaborative editing.** CRDT-based multi-user conflict resolution extending the editor module (§19).
6. **SNMP and streaming telemetry adapters.** Community adapters for SNMP, Napalm, and gNMI.
7. **Parallel edge animation.** Per-link traffic flow animation within a LAG bundle.

References

[1] NetBox Labs, *NetBox: The leading solution for network automation*, 2024. [Online]. Available: https://netbox.dev

[2] E. R. Gansner and S. C. North, "An open graph visualization system and its applications to software engineering," 11, vol. 30, 2000, pp. 1203–1233.

[3] M. Chimani, C. Gutwenger, M. Jünger, G. W. Klau, K. Klein, and P. Mutzel, "The open graph drawing framework (OGDF)," *Handbook of Graph Drawing and Visualization*, pp. 543–569, 2013.

[4] M. Bostock, V. Ogievetsky, and J. Heer, "D3: Data-driven documents," *IEEE Transactions on Visualization and Computer Graphics*, vol. 17, no. 12, pp. 2301–2309, 2011.

[5] yWorks GmbH, *yEd graph editor*, 2024. [Online]. Available: https://www.yworks.com/products/yed

[6] bluss, *Petgraph – graph data structure library for Rust*, 2024. [Online]. Available: https://crates.io/crates/petgraph

[7] K. Sugiyama, S. Tagawa, and M. Toda, "Methods for visual understanding of hierarchical system structures," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 11, no. 2, pp. 109–125, 1981.

[8] T. M. J. Fruchterman and E. M. Reingold, "Graph drawing by force-directed placement," *Software — Practice and Experience*, vol. 21, no. 11, pp. 1129–1164, 1991.

[9] L. Verlet, "Computer "experiments" on classical fluids. I. thermodynamical properties of Lennard-Jones molecules," *Physical Review*, vol. 159, pp. 98–103, 1967.

[10] J. Barnes and P. Hut, "A hierarchical $O(N \log N)$ force-calculation algorithm," *Nature*, vol. 324, pp. 446–449, 1986.

[11] J. Christensen, J. Marks, and S. Shieber, "An empirical study of algorithms for point-feature label placement," in *ACM Transactions on Graphics*, vol. 14, 1995, pp. 203–232.

[12] N. Beckmann, H.-P. Kriegel, R. Schneider, and B. Seeger, "The R*-tree: An efficient and robust access method for points and rectangles," *ACM SIGMOD Record*, vol. 19, no. 2, pp. 322–331, 1990.

[13] Stoeoef, *Rstar – R*-tree spatial index for Rust*, 2024. [Online]. Available: https://crates.io/crates/rstar

[14] D. Holten and J. J. van Wijk, "Force-directed edge bundling for graph visualization," *Computer Graphics Forum*, vol. 28, no. 3, pp. 983–990, 2009.

[15] H. C. Purchase, "Metrics for graph drawing aesthetics," *Journal of Visual Languages and Computing*, vol. 13, no. 5, pp. 501–516, 2002.

[16] C. Dunne and B. Shneiderman, "Motif simplification: Improving network visualization readability with fan, connector, and clique glyphs," *Proc. ACM CHI*, pp. 3247–3256, 2015.

[17] J. Abello, F. van Ham, and N. Krishnan, "ASK-GraphView: A large scale graph visualization system," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 12, 2006, pp. 669–676.

[18] A. Haas et al., "Bringing the web up to speed with WebAssembly," in *Proc. ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2017, pp. 185–200.

References

[1] NetBox Labs, *NetBox: The leading solution for network automation*, 2024. [Online]. Available: https://netbox.dev

[2] E. R. Gansner and S. C. North, "An open graph visualization system and its applications to software engineering," 11, vol. 30, 2000, pp. 1203–1233.

[3] M. Chimani, C. Gutwenger, M. Jünger, G. W. Klau, K. Klein, and P. Mutzel, "The open graph drawing framework (OGDF)," *Handbook of Graph Drawing and Visualization*, pp. 543–569, 2013.

[4] M. Bostock, V. Ogievetsky, and J. Heer, "D3: Data-driven documents," *IEEE Transactions on Visualization and Computer Graphics*, vol. 17, no. 12, pp. 2301–2309, 2011.

[5] yWorks GmbH, *yEd graph editor*, 2024. [Online]. Available: https://www.yworks.com/products/yed

[6] bluss, *Petgraph – graph data structure library for Rust*, 2024. [Online]. Available: https://crates.io/crates/petgraph
[7] K. Sugiyama, S. Tagawa, and M. Toda, "Methods for visual understanding of hierarchical system structures," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 11, no. 2, pp. 109–125, 1981.
[8] T. M. J. Fruchterman and E. M. Reingold, "Graph drawing by force-directed placement," *Software — Practice and Experience*, vol. 21, no. 11, pp. 1129–1164, 1991.
[9] L. Verlet, "Computer "experiments" on classical fluids. I. thermodynamical properties of Lennard-Jones molecules," *Physical Review*, vol. 159, pp. 98–103, 1967.
[10] J. Barnes and P. Hut, "A hierarchical $O(N \log N)$ force-calculation algorithm," *Nature*, vol. 324, pp. 446–449, 1986.
[11] J. Christensen, J. Marks, and S. Shieber, "An empirical study of algorithms for point-feature label placement," in *ACM Transactions on Graphics*, vol. 14, 1995, pp. 203–232.
[12] N. Beckmann, H.-P. Kriegel, R. Schneider, and B. Seeger, "The R*-tree: An efficient and robust access method for points and rectangles," *ACM SIGMOD Record*, vol. 19, no. 2, pp. 322–331, 1990.
[13] Stoeoef, *Rstar – R*-tree spatial index for Rust*, 2024. [Online]. Available: https://crates.io/crates/rstar
[14] D. Holten and J. J. van Wijk, "Force-directed edge bundling for graph visualization," *Computer Graphics Forum*, vol. 28, no. 3, pp. 983–990, 2009.
[15] H. C. Purchase, "Metrics for graph drawing aesthetics," *Journal of Visual Languages and Computing*, vol. 13, no. 5, pp. 501–516, 2002.
[16] C. Dunne and B. Shneiderman, "Motif simplification: Improving network visualization readability with fan, connector, and clique glyphs," *Proc. ACM CHI*, pp. 3247–3256, 2015.
[17] J. Abello, F. van Ham, and N. Krishnan, "ASK-GraphView: A large scale graph visualization system," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 12, 2006, pp. 669–676.
[18] A. Haas et al., "Bringing the web up to speed with WebAssembly," in *Proc. ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2017, pp. 185–200.

A Topology YAML Schema

Listing 1. Full YAML topology schema

```yaml
name: string              # Required: topology name
render:                   # Optional: rendering hints
  preset: string          #   Layout preset (default|sugiyama|radial|geo|subway)
  layout: string          #   Layout algorithm (force_directed|hierarchical|
                          #     radial|geographic|isometric|orthogonal|subway|tree)
  theme: string           #   Theme (network|dark|light|blueprint|datacenter|print|high-contrast)
  width: number           #   Canvas width in pixels
  height: number          #   Canvas height in pixels
  lod: string             #   LOD profile (none|conservative|balanced|aggressive|auto)
  seed: number            #   RNG seed for deterministic layout
  refiners: [string]      #   Refiner chain override
  metric_encoding:        #   Data-driven visual encoding
    - attribute: string   #     Node attribute name
      channel: string     #     fill_color|size|opacity|edge_color
      palette: string     #     blues|greens|oranges|reds|purples|greys
      min: number
      max: number
nodes:                    # Required: list of nodes
  - id: string            #   Unique node identifier
    label: string         #   Display label (defaults to id)
    node_type: string     #   Device type (router|switch|server|firewall|cloud|
                          #     l3_switch|load_balancer|database|endpoint|wireless_ap)
    attributes: map       #   Arbitrary key-value metadata
    position: { x, y }    #   Position hint (optional)
    group: string         #   Group membership
    layer: string         #   Layer membership (isometric)
    highlight: string     #   Effect name or true for default glow
    lat: number           #   Geographic latitude (geographic layout)
    lon: number           #   Geographic longitude (geographic layout)
    interfaces:           #   Interface list
      - id: string
        label: string
        status: string    #     up|down|admin_down|errors
        mode: string      #     dot|port|detailed
        marker: string    #     none|dot|square|diamond
```

```
    constraints:          #   Layout constraints
      pin: { x, y }       #     Absolute position
      align_with: string  #     node ID (same Y)
      below: string       #     node ID
      min_separation: number
edges:                    # Required: list of edges
  - from: string          #   Source node ID
    to: string            #   Target node ID
    id: string            #   Explicit edge ID
    label: string         #   Edge label
    weight: number        #   Edge weight (default: 1.0)
    directed: boolean     #   Default: false
    bond_group: string    #   Bond/LAG group ID
    parallel_mode: string #   offset|bundled|bonding_icon|stacked|angled
    gradient:
      start_color: string #   hex color at source
      end_color: string   #   hex color at target
    attributes: map
groups:                   # Optional: group definitions
  - { id: string, label: string, style: map }
layers:                   # Optional: isometric layer definitions
  - { id: string, label: string, order: number }
paths:                    # Optional: path emphasis definitions
  - id: string
    label: string
    emphasis: string      #   primary|secondary
    fanout_mode: string   #   fixed|scaled|hybrid
    nodes: [string]
traffic:                  # Optional: traffic flow animation
  edges:
    - id: string
      utilization: number #   0.0--1.0
      direction: string   #   forward|reverse|bidirectional
      style: string       #   dot|dash
annotations:              # Optional: SVG annotation markup
  - anchor: string
    type: string          #   text|callout
    text: string
    shape: string         #   circle|box|arrow
    layer: string
```

B Configuration Reference

B.1 CLI Flags

B.1.1 Input and Output

Flag	Type	Default	Description
`--input`, `-i`	path	(required)	Topology file; globs and - for stdin
`--output`, `-o`	path	auto	Output file path
`--config`	path	—	JSON config file
`--format`	string	`svg`	Output format: svg, png, pdf, html, dot
`--dpi`	integer	96	DPI for PNG export

B.1.2 Layout

Flag	Default	Description
`--layout`	auto	Algorithm: force-directed, hierarchical, radial, isometric, geographic, orthogonal, subway, tree
`--preset`	—	Layout preset: default, wan, datacenter, sugiyama, geographic, subway, …
`--template`	—	Topology template: spine-leaf, hub-spoke, ring, multi-tier
`--seed`	42	RNG seed for deterministic layout

B.1.3 Appearance

Flag	Default	Description
`--theme`	`network`	Theme (7 built-in): light, dark, network, blueprint, datacenter, print, high-contrast
`--labels`	`label`	Label mode: off, label, id
`--node-size`	auto	Node radius in scene units
`--lod`	auto	LOD profile: none, conservative, balanced, aggressive, auto

B.1.4 Canvas and Viewport

Flag	Default	Description
`--width`	1200	SVG viewport width (px)
`--height`	800	SVG viewport height (px)
`--padding`	40	ViewBox padding (px)
`--padding-ratio`	0.06	Padding as fraction of scene

B.1.5 Force-Directed Tuning

Flag	Default	Description
`--link-distance`	30.0	Target link length (scene units)
`--link-strength`	auto	Edge spring strength (0.0–1.0)
`--charge-strength`	−30.0	Many-body charge (negative = repel)
`--max-iterations`	300	Maximum simulation iterations
`--soft-anchor-strength`	0.0	Anchor strength for hinted nodes
`--refine`	false	Apply post-layout edge refinement

B.1.6 Geographic Layout

Flag	Default	Description
`--show-map`	false	Show background coastline map
`--great-circle-arcs`	false	Render edges as curved arcs
`--show-distances`	false	Show geodesic distance labels
`--distance-unit`	km	Distance unit: km or mi

B.1.7 Filtering

Flag	Description
`--show-type`	Only render nodes of these types (comma-separated)
`--hide-type`	Hide nodes of these types
`--show-tag`	Only render nodes with these tags
`--hide-tag`	Hide nodes with any of these tags
`--show-group`	Only render nodes in these groups
`--hide-group`	Hide nodes in these groups
`--filter-file`	Load filter rules from YAML file
`--path`	BFS path query between two node IDs
`--highlight-path`	Highlight a named path (repeatable)

B.1.8 Analysis Features

Flag	Description
`--diff <path>`	Compare against a base topology and visualize changes
`--traffic <path>`	Traffic overlay configuration YAML
`--annotations <path>`	Annotation markup YAML
`--hide-layers`	Suppress named annotation layers
`--show-layers`	Force-show named annotation layers
`--cluster`	Enable automatic community detection
`--algorithm`	Clustering algorithm: louvain, label-propagation, edge-betweenness
`--metric <field>`	Map a node attribute to color
`--palette <name>`	Color palette: blues, greens, oranges, reds, purples, greys
`--analytics`	Compute and display graph analytics

B.1.9 Quality and Diagnostics

Flag	Default	Description
`--quality-threshold`	85.0	Quality score warning threshold
`--fail-on-low-quality`	false	Exit with error if below threshold
`--deterministic`	true	Deterministic output mode
`--diagnostics`	—	Structured diagnostics: json, yaml, ndjson
`--verbose, -v`	false	Show debug statistics

B.2 CLI Subcommands

Command	Description
`list-examples`	List built-in topology examples
`print-example <name>`	Print a built-in topology to stdout
`list-templates`	List layout templates
`list-adapters`	List data source adapters
`import`	Import topology from adapter into YAML
`schema`	Print JSON Schema (config, topology, or diagnostics)
`print-config <name>`	Print an example config to stdout
`validate`	Validate topology/config against schema
`migrate`	Normalize files to current schema version
`check-quality`	Analyze SVG files for quality issues
`diff`	Compare two topologies, visualize changes
`show`	Render topology to ASCII art in the terminal

B.3 JSON Configuration File

NetVisConfig JSON

```
{
  "schema_version": "1.0",
  "layout": "force_directed",
  "quality": "balanced",
  "width": 1200.0,
  "height": 800.0,
  "effects": { "max_filtered_elements": 50 }
}
```

Key	Type	Default	Description
`layout`	string	`force_directed`	Layout algorithm
`quality`	string	`balanced`	Preset: quick (0.33x), balanced (1.0x), quality (3.0x)
`width`	float	800.0	Viewport width
`height`	float	600.0	Viewport height
`effects.max_filtered_elements`	int	50	SVG filter budget

Table 14. JSON configuration keys.

Index

List of Figures

List of Tables

List of Design Decisions & Rules

Revision History

Version	Date	Changes
1.5.0	March 2026	Subway layout; TreeLayout; interface rendering; node shape library; path emphasis; metric encoding; edge gradients; additional refiners (Starburst, Deoverlap, EdgeClipping, SmartLabel, LabelNudging, BoundaryBundling); terminal renderer
1.4.0	March 2026	Parallel edges (5 modes); Bond group/LAG aggregation; Orthogonal layout; Clarity scoring; Graph analytics; Filter DSL; Fuzzy validation
1.3.0	March 2026	Margin Insights; Cover Diagrams; Design Rule Primitives
1.2.0	March 2026	Integrated W2P Architectural Strategy; Formalized Type Logic
1.1.0	March 2026	Routing refiners, topology diffing, timeline, traffic animation, annotations, interactive editor, visual effects, 16 quality metrics, configuration reference
1.0.0	Jan 2025	Initial Formal Pipeline Specification